



MADRAS SECURITY PRINTERS
The Future Looks Smart

System Requirement Specification (SRS)

New Web-based Portal and Mobile-based Application for Bihar Building and other Construction Workers Welfare Board, Department of Labour Resources, GoB.

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About Madras Security Printers

Established in 1976, our organization is a vital component of the esteemed Eagle Corporate Group, headquartered in Chennai, India, and operating offices across four continents. As one of the world's leading high-security solution providers, MSP boasts an impressive array of certifications and standards, including ISO 9001, ISO 20000, ISO 27001, CMMI Level 5, VISA, MASTERCARD, RUPAY, SCOSTA, and OHASIS 18001.

Pioneering system integration since 2006, we have consistently demonstrated our commitment to excellence. A noteworthy aspect of our operations is the significant percentage of our revenues dedicated to corporate social responsibility (CSR).

MSP has earned recognition from prestigious bodies such as ICMA, IHMA, and ASPA, and has received accolades from the Tax Stamp Forum and e-India. Our commitment to innovation and excellence is further highlighted by our engagement in smart and safe city projects, with ongoing execution in prominent locations such as Mangalore Smart City, Karnal Smart City, Karnataka Municipal Data Society, Chennai Safe City, and Thiruvananthapuram Smart City, Pondicherry Police Station Surveillance System. Additionally, MSP has taken the lead in pioneering Data Centre experience projects, City Surveillance Projects and extensive experience in e-Governance initiatives. Our commitment to excellence and innovation is underscored by these significant achievements.

Introduction

The **BBOCWW Board** is dedicated to the welfare of building and construction workers, ensuring their rights and well-being through various initiatives. As of now, approximately **34 lakh workers** are registered with the Board. To support these workers, the Board operates **sixteen welfare schemes**, providing financial assistance, healthcare benefits, skill development programs, and other essential services. The funding for these initiatives comes from the **CESS collected from construction activities** across the state.

During the **financial year 2023-24**, the Board successfully provided benefits to **1,09,638 registered workers** through its existing online portal. However, the current system faces multiple challenges and lacks several critical features needed for efficient service delivery.

Key Challenges with the Existing System:

1. **Lack of Advanced Biometric Authentication** – The current portal does not have an intelligent, robust mechanism to biometrically verify beneficiaries. As a result, redundant and duplicate beneficiaries remain in the system, leading to inefficiencies and potential misuse of resources.
2. **Cumbersome Approval Process** – The system does not facilitate a seamless and automated approval workflow, making the process slow and prone to manual errors.
3. **Limited Monitoring & Reporting Capabilities** – The Board currently has restricted options to track **enrolment status and fund disbursement** effectively. A modern, automated dashboard with built-in **intelligent design, data analytics, and report generation capabilities** is essential for better decision-making.
4. **Performance Issues** – The existing system struggles with operational inefficiencies, affecting its ability to handle large-scale data processing and timely disbursement of benefits.

To enhance service delivery and ensure transparency, the Board requires a **modernized, intelligent system** that integrates biometric authentication, streamlines approvals, and provides real-time insights through an advanced dashboard. Upgrading the existing infrastructure will enable the Board to efficiently manage welfare schemes and maximize the impact of its initiatives for construction workers.

Project Purpose & Objectives

Project Purpose

The primary objective of this project is to **design, develop, deploy and maintain** a comprehensive **web portal and mobile application** that will serve as a centralized digital platform for the **registration of construction workers**, seamless **implementation of welfare schemes**, and efficient **collection of cess** from contractors, owners, and employers. This initiative aims to enhance transparency, improve service delivery, and streamline operations related to labour welfare and compliance.

By leveraging modern technology, the platform will ensure **faster processing, better accessibility, data accuracy, and real-time monitoring** of all key activities, enabling the Board to provide improved support to the construction workforce.

Key Objectives of the Project

Construction Worker Registration & Management

- ✚ Enable hassle-free **registration of construction workers** through both web and mobile applications.
- ✚ Provide digital **renewal options** for registered workers to maintain continuous access to welfare benefits.
- ✚ Facilitate **real-time updating of worker details**, including employment status,

contact information, and eligibility for various schemes.

Welfare Scheme Implementation & Processing

- ✚ Allow workers to **apply for welfare schemes online** with minimal manual intervention.
- ✚ Implement a **transparent and automated approval process**, ensuring quick and fair disbursement of benefits.
- ✚ Maintain a **centralized database of scheme beneficiaries** for accurate monitoring and reporting.

Construction & Establishment Registration

- ✚ Enable the **registration of construction and building projects/establishments** by contractors, employers, and owners.
- ✚ Ensure compliance by making it mandatory for employers to **register their projects** and workers through the system.

Cess Collection, Calculation & Monitoring

- ✚ **Automate cess collection** from contractors, project owners, and employers.
- ✚ Implement a system for **accurate cess calculation** based on project details and applicable regulations.
- ✚ Include provisions for **interest calculation on delayed payments** to ensure compliance and timely revenue collection.

Management Information System (MIS) & Analytics

- ✚ Develop a **comprehensive MIS dashboard** for real-time tracking of worker registrations, scheme applications, cess collection, and fund disbursement.
- ✚ Enable **data-driven decision-making** through automated **report generation**,

data mining, and analytical tools.

- ✚ Provide **customizable reports** in various formats to meet administrative and policy needs.

Expected Outcomes

- **Increased efficiency** in worker registration and welfare scheme management.
- **Elimination of fraudulent registrations** through biometric authentication and smart validation techniques.
- **Faster processing and approval** of welfare applications with minimal paperwork.
- **Real-time insights** into labour welfare operations, scheme implementation, and financial transactions.
- **Improved compliance and revenue generation** through automated cess collection and monitoring.
- **Enhanced user experience** through mobile-friendly access and multilingual support for workers.

This initiative will **transform the Board's welfare service delivery** by creating an integrated, user-friendly, and efficient platform that benefits both workers and the administration.

Scope of Project Work

Overview

The scope of this project involves the **design, development, implementation, and maintenance** of a **web portal and mobile application** to facilitate the **registration of**

construction workers, management of **welfare schemes**, and **collection of cess** from contractors, employers, and owners.

This digital platform will ensure **seamless service delivery**, **enhanced transparency**, and **improved efficiency** in labour welfare operations by incorporating modern technology-driven solutions such as **biometric authentication**, **real-time monitoring**, **automated approvals**, **data analytics**, and **intelligent dashboards**.

Key Features & Functionalities

Worker Registration & Management

- ✚ **Secure Registration & Login:** Workers will be able to register online using a **biometric authentication** system for secure and accurate verification.
- ✚ **Online Scheme Application:** Workers can apply for **welfare schemes** digitally after bio-authentication.
- ✚ **Renewal & Updating:** Facility for workers to **renew their registration and update their details** online through biometric validation.
- ✚ **Real-Time Tracking:** Applicants can track their **registration and scheme application status** in real-time.

Resource Centre (CMS - Content Management System)

- ✚ **Details of welfare schemes** and eligibility criteria.
- ✚ **Legal Acts & regulations** governing construction workers' welfare.
- ✚ **Training materials, safety guidelines, and best practices.**
- ✚ **Ready Reckoner** for frequently asked questions and quick references.

Feedback & Grievance Management

- ✚ **Feedback Mechanism:** Users can submit suggestions and report issues regarding portal functionalities.

- ✚ **Grievance Redressal:** A **ticket-based system** will allow workers and stakeholders to report concerns and track resolution progress.
- ✚ **Dedicated Helpdesk:** A **support team** will be available to assist workers and resolve grievances effectively over a **5-year period**.

Cess Collection & Processing

- ✚ **Online Cess Payment:** Employers, contractors, and project owners can register **construction projects and buildings** and submit required cess payments.
- ✚ **Automated Calculation:** The system will **automatically compute cess and applicable interest** based on project details.
- ✚ **Self-Registration for Contractors:** Contractors can **self-register online** and declare their construction activities for **compliance and cess submission**.

Integration with External Systems

The platform will include **API-based integrations** with:

- ✚ **e-Shram Portal (Government of India)** for data exchange and worker verification.
- ✚ **NACH (National Automated Clearing House)** for seamless digital payments.
- ✚ **SMS & Email Gateway** for instant notifications and communication.
- ✚ **Aadhaar-based Authentication** to validate worker identity and prevent duplication.

Management Information System (MIS) & Business Intelligence (BI) Dashboard

A **real-time, data-driven dashboard** will provide comprehensive analytics and insights:

- ✚ **Dashboard Overview:** Displays key statistics like **total registered workers, total cess collected, pending applications, and scheme disbursements**.
- ✚ **Geographic Heat Maps:** Visual representation of **worker and contractor distribution** at different administrative levels (state, district, block, ULB, Panchayat).

- ✚ **Trend Analysis:** Tracks **growth trends in registrations, cess payments, and scheme utilization** over time.
- ✚ **Drill-Down Reports:** Enables stakeholders to analyse **district/block-wise data** for better decision-making.

Data Migration & Maintenance

- ✚ **Seamless Migration of Existing Data** from the legacy system to the new platform.
- ✚ **Continuous Maintenance & Support** for **5 years post go-live** to ensure smooth operation and updates.
- ✚ **Training & Handholding Support** to help stakeholders effectively use the platform.

Additional Key Features of the Portal & Mobile App

Organizational & Public Information

- ✚ **'About Us' Section:** Highlights the Board's **mission, objectives, and contributions**.
- ✚ **Organizational Structure (Organogram):** Graphical representation of **roles and responsibilities** within the Board.
- ✚ **Right to Information (RTI):** Dedicated section to ensure **public transparency and accountability**.

Notifications & Public Communication

- ✚ **Grievance Redressal:** Ticket-based system for **issue reporting and resolution**.
- ✚ **Feedback Mechanism:** Users can submit **portal improvement suggestions**.
- ✚ **Media Section:** Hosts **government notifications, news updates, and announcements**.
- ✚ **Tenders Section:** Displays **open tenders for procurement and services**.

User Experience & Accessibility Enhancements

- ✚ **Aadhaar, SMS, and Payment Gateway Integrations.**

- ✚ **Advanced Search Functionality:** Provides **quick access to curated information**.
- ✚ **Bilingual Support:** The portal and mobile app will be available in **Hindi and English**.
- ✚ **Interactive Chat box:** Assistant for **FAQs and instant responses in Hindi & English**.
- ✚ **Downloadable Content:** Scheme-related documents and forms will be available for **download in Hindi & English**.
- ✚ **Mobile App Availability:** The mobile application will be **accessible on Google Play Store** for easy installation.
- ✚ **Community-Based Forum:** Category-based **Q&A platform** for worker interactions and discussions.
- ✚ **Cybersecurity Compliance:** The system will adhere to **government-mandated cybersecurity protocols** for **data protection and service reliability**.

Project Approach

Project Execution Plan

The successful implementation of this project will follow a structured approach covering **requirement gathering, system design, development, testing, deployment, and training**. This approach ensures that the web portal and mobile application are developed with **scalability, security, and user-friendliness** in mind while integrating with critical external systems like **e-Shram, Aadhaar, and payment gateways**.

Requirement Gathering and Analysis

Stakeholder Engagement:

- ✚ Conduct **workshops, interviews, and focus group discussions** with key stakeholders, including:
 - **BBOCWW Board officials** (policy and decision-makers).
 - **Technical experts** for system integration and security compliance.
 - **Government representatives** for regulatory alignment.

- **Construction workers and contractors** for end-user insights.

✚ Document pain points, process inefficiencies, and requirements to ensure the new system addresses all critical needs.

Functional and Non-Functional Requirements:

✚ Identify and document:

- **User stories** for each type of user (workers, Board officials, contractors, employers).
- **Core functionalities** (registration, scheme application, cess collection, document uploads, feedback mechanism, etc.).
- **Non-functional expectations**, including **performance, scalability, data security, and response time benchmarks**.

Technical Requirements:

✚ Define **API integration points** with:

- **e-Shram** (for worker data synchronization).
- **Aadhaar authentication** (for biometric validation).
- **NACH Payment Gateway** (for financial transactions).
- **SMS and Email services** (for notifications).

✚ Assess third-party tools and services required for:

- **BI-powered dashboards** for analytics.
- **Document and content management systems** for administrative efficiency.

Deliverable:

✚ **Detailed System Requirement Specification (SRS) document**, outlining user needs, system architecture, security protocols, and integration mechanisms.

System Design

High-Level Architecture:

- ✚ Develop a **scalable, high-availability architecture** to support **thousands of concurrent users**.
- ✚ Implement **deployment** with **data backup and disaster recovery provisions**.
- ✚ Ensure **secure access management** through **multi-factor authentication (MFA)** for Board administrators.

Database Design:

- ✚ Design an optimized **relational database (PostgreSQL)** to store:
 - **Worker registrations and renewals.**
 - **Scheme applications and approvals.**
 - **Cess payment records.**
 - **Real-time application tracking and notifications.**
- ✚ Include **document management capabilities** for the secure storage of submitted forms and worker credentials.

API Design:

- ✚ Develop APIs for:
 - **Aadhaar-based biometric authentication.**
 - **Data exchange with e-Shram.**
 - **Real-time payment processing and cess validation.**

User Interface (UI/UX) Design:

- ✚ Develop **interactive wireframes and prototypes** for the **web portal and mobile app**.
- ✚ Ensure:
 - **User-friendly interfaces** for workers with limited digital literacy.
 - **Bilingual support (Hindi & English).**

- **Mobile responsiveness and accessibility compliance.**

Deliverables:

- + **Technical Design Document (TDD)** outlining system components, workflows, and data structures.
- + **Prototype wireframes** for web and mobile applications.

Development Phase

Project Management Approach:

- + Utilize **CMMI Level-5 best practices** for structured project execution.
- + Follow an **iterative development methodology** to allow **continuous testing and stakeholder feedback**.

Backend Development:

- + Develop a **robust backend using Java/Spring Boot**, focusing on:
 - Worker registration, renewal, and scheme application processing.
 - Secure cess payment and tracking system.
 - Contractor and employer registration and compliance validation.
 - API development for third-party integrations.

Frontend Development:

- + Develop:
 - **Web portal using Angular** for a modern, responsive UI.
 - **Mobile app (Android)** for **on-the-go access** to services.
- + Ensure **cross-device compatibility and biometric login functionality**.

Business Intelligence (BI) Dashboard:

- + Integrate an **open-source BI platform** for **real-time analytics and decision-making**.

- ✚ Enable features like **geographic heat maps, trend analysis, and drill-down reporting.**

Data Migration:

- ✚ Develop **migration scripts** to seamlessly transfer **legacy data** to the new system **without data loss or corruption.**

Deliverables:

- ✚ **Fully functional web portal and mobile application** with integrated features.

Quality Assurance and Testing

Unit Testing:

- ✚ Test individual modules (**registration, payments, scheme applications**) for **functionality and performance.**

Integration Testing:

- ✚ Validate seamless integration between **biometric authentication, payment gateways, e-Shram API, and document management systems.**

User Acceptance Testing (UAT):

- ✚ Conduct **hands-on testing** with **Board officials, contractors, and registered workers** to ensure **real-world usability.**

Security Testing:

- ✚ Perform **penetration testing, vulnerability assessments, and compliance audits** to ensure data security and protection against cyber threats.

Performance Testing:

- ✚ Conduct **load testing** using industry-standard tools to ensure the system can handle **high user traffic and concurrent transactions.**

Deliverables:

- ✚ **Test Plan, Test Reports, Security Audit Reports, and UAT Sign-Off.**

Deployment and Go-Live

Deployment Process:

- ✚ Deploy the web portal and mobile app to the **production environment** with **zero downtime migration**.
- ✚ Implement **monitoring and logging tools** to ensure **real-time performance tracking**.

Mobile App Store Launch:

- ✚ Publish the **mobile application on Google Play Store** and ensure compliance with app store guidelines.

Deliverables:

- ✚ **Live, fully functional web portal and mobile app.**

Training and Knowledge Transfer

Training Programs:

- ✚ Conduct training sessions for:
 - **Board officials** to manage registrations, approvals, and reports.
 - **Contractors and employers** on cess compliance and project registration.
 - **Construction workers** on scheme applications and tracking benefits.

Knowledge Transfer:

- ✚ Develop **detailed user manuals, video tutorials, and FAQs**.
- ✚ Set up a **dedicated online helpdesk and chatbox** for continuous user support.

Deliverables:

- ✚ **Training materials, video tutorials, and user manuals.**

References

SL No	Reference Documents
1	Scope document/RFP
2	Existing previous system screenshots
3	BBOCWB Rule book & ACTs hosted in present website
4	MOM of System Study
5	Collected sample formats
6	Questionnaire

Overall Description

Product Perspective

The proposed application is intended to provide a centralized and web enabled MIS application to facilitate the BBCLWB for Worker registration, scheme management, sanction order generation and payment to Worker. It will facilitate the users of block, district and state offices for better monitoring of the scheme management and tracking of payment process. The communication process will be enhanced so that the request for approval process will be simpler and the delay in file processing will be reduced.

The application will be a centralized web application designed and developed using the three-layer architecture. Three-layer architecture is a client-server architecture in which the functional processes of user interface, business logic and data access layer are developed and maintained in separate layer. The three-layer architecture is a software design pattern and well-established software architecture.

Presentation Layer. Responsibility for presentation and user interaction resides with the presentation layer. These client components enable the user to interact with the business layer in a secure manner. The information is passed and retrieve to the presentation layer with the help of business layer.

Business Layer. The business layers are commonly referred to as the "application logic layer". These processes manage the business logic of the application, and are permitted access to the data layer. The application logic layer is where most of the processing work occurs. This layer encapsulates the business functionalities of the modules. It interacts between presentation layer and data access layer.

Data Access Layer. The data accesses layer is connected to the database. The data interaction between the application and database occurs through the data access layer. The data access layer interacts with database and business layer.

Product Features

Key Features and Benefits

The proposed **web and mobile application** aim to **fully automate** the end-to-end process of **worker registration, scheme management, and cess collection** from contractors. This centralized digital solution will be accessible via **web and mobile platforms**, allowing seamless interaction between various offices across different locations.

Key Features and Benefits

Secure and Verified Registration Process

- ✚ The application integrates **advanced biometric authentication, OTP verification, and Aadhaar-based UIDAI authentication** to ensure that only **genuine workers and beneficiaries** are registered.

- ✚ Eliminates **duplicate registrations** of workers and contractors, ensuring that welfare benefits reach the right individuals.

Intelligent Scheme Management and Decision Support

- ✚ The system includes a **built-in business logic engine** that automates **eligibility verification and sanctioning processes** for different welfare schemes.
- ✚ Workers can apply for **any of the 16 schemes**, and the system will automatically classify applications based on **scheme criteria and worker eligibility**.
- ✚ The **approval process is dynamic and streamlined**, reducing delays in sanctioning benefits.

Real-Time Tracking and Monitoring

- ✚ **Board officials can track applications in real-time**, ensuring complete visibility into **pending, approved, and rejected applications**.
- ✚ The system maintains an **audit trail of all scheme approvals and cess payments**, ensuring transparency.
- ✚ **Workers can track their application status** through the portal or mobile app.

Contractor Cess Collection and Compliance Management

- ✚ Contractors can easily **register their projects and process cess payments online**, ensuring compliance with labour welfare regulations.
- ✚ The board can **monitor cess collection**, track defaulters, and generate **automated reminders** for pending payments.

Professional Content & Document Management System

- ✚ The system will maintain an **interactive, up-to-date website and mobile app** using a **robust content and document management system (CMS/DMS)**.
- ✚ Provides **easy access to welfare scheme details, forms, policies, and regulations** for workers, contractors, and Board officials.

Enhanced Grievance Redressal Mechanism

- ✚ Workers can **raise grievances through a ticket-based helpdesk system**, ensuring proper tracking and resolution of complaints.
- ✚ A **dedicated support system** will ensure that **queries and complaints** are addressed efficiently.

Interactive Chatbot & User Assistance

- ✚ The system will feature a **chatbox** to assist workers in **understanding scheme eligibility, tracking applications, and resolving queries**.
- ✚ This **self-service feature** makes it easier for workers to interact with the system, reducing dependency on physical office visits.

Seamless Application Processing & Payment Tracking

- ✚ The application ensures **systematic arrangement and categorization** of worker applications based on scheme eligibility criteria.
- ✚ Board officials can **easily track disbursements** and payments made across different schemes.
- ✚ Workers receive **timely notifications** about their benefits and approvals.
- ✚ The proposed **web and mobile-enabled solution** will **revolutionize worker welfare management** by ensuring:
 - ✓ **Secure, verified registration** of workers and contractors.
 - ✓ **Efficient tracking and approval** of scheme applications.
 - ✓ **Automated cess collection and compliance monitoring** for contractors.
 - ✓ **A professional, content-driven portal** for all stakeholders.
 - ✓ **Seamless grievance handling** and self-service options for workers.
 - ✓ **Real-time monitoring & reporting** to enhance decision-making and transparency.

This system will significantly **enhance service delivery, ensure compliance, and improve worker welfare administration** through **automation, intelligence, and transparency**.

Members

SL No	Members	Actors	Functionality / Cases
1	Common Citizen	Web_App	Website & Mobile App Front End for Public
2	HQ Admin	Web Admin	Website Content & Document Management System
3	HQ Admin	HQ Module	Headquarters Admin Module
4	HQ Admin	District Login	Under HQ District Workflow Can Be Managed By HQ
5	HQ Admin	Block Login	Under HQ Block Workflow Can Be Managed By HQ
6	Worker / Labour	Labour Module	Applicant/Worker/Labour Workflow Management
7	Contractor / Employer	Employer Module	Contractors/ Building Owner / Employer Workflow Management
8	Officer Login	Leo Module	Labour Enforcement Officer Workflow Management
9	Officer Login	Ls Module	Labour Supervenient Workflow Management
10	Officer Login	DLC Module	Deputy Labour Commissioner Workflow Management

Operating Environment & Architecture

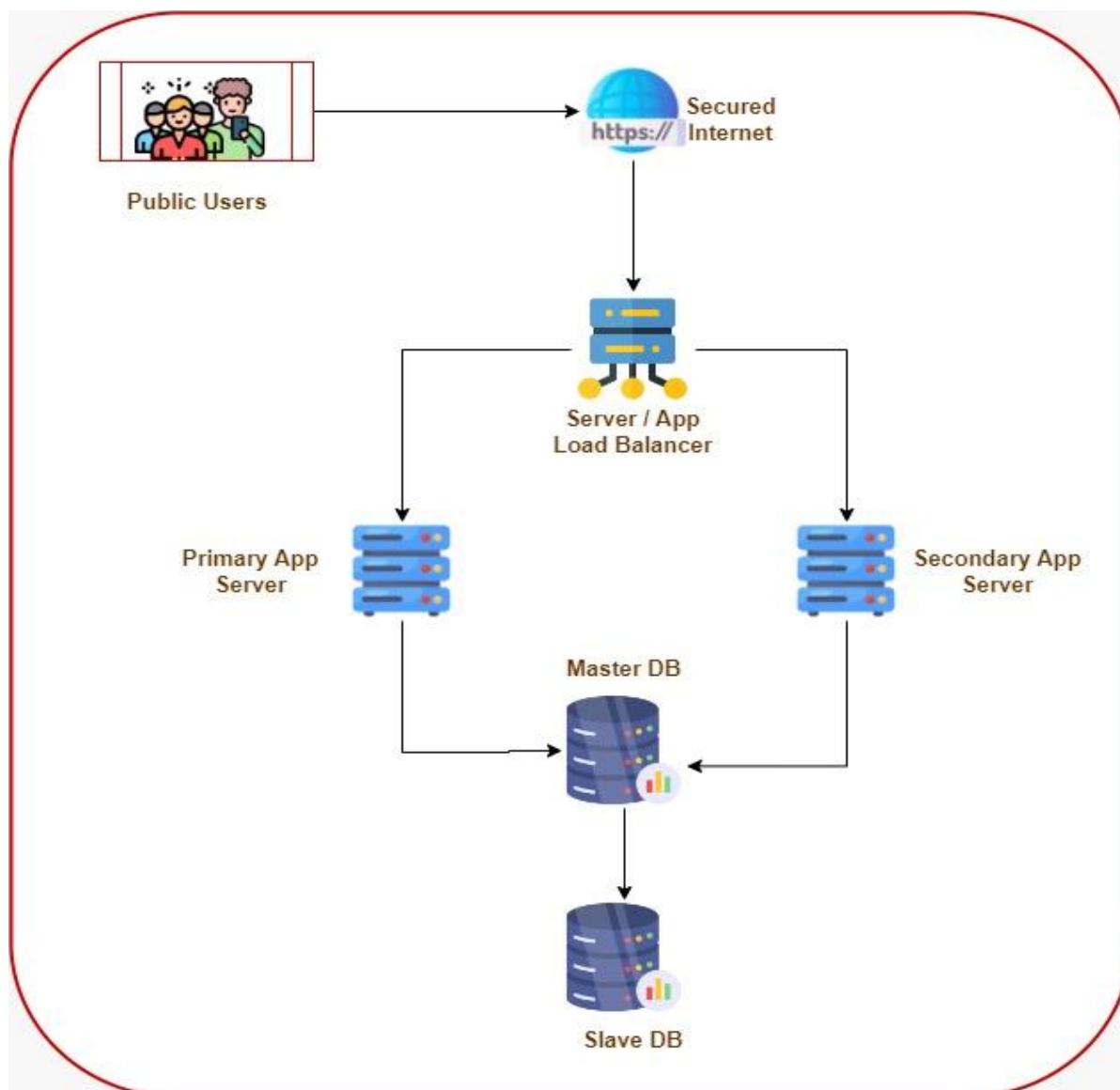
Software Requirement

Sl. No	Category	Description
1	Operating system	Ubuntu Stable Version
2	Application Framework	Front End: Angular. Language: typescript Java with Spring boot
3	Database	Postgres Stable Version
4	Web Server	Nginx server with Tomcat
5	Web browsers	Mozilla Firefox 21.0,22,0 Google Chrome 52.0.2743.116 Safari
6	Domain , SSL	The Nodal agency will provide the Domain name for hosting the application, along with SSL certification and VPN access for four users to securely access the server.
7	Mobile App	Flutter Stable Version
8	App hosting account authorization	The department will provide access to the Google Play Store account.
9	Third party integration Document	UIDAI Adhaar API NACH API payment document & access SMSTP & SMS gateway document & access

Hardware Requirement (Sizing may change based on migration of data from current running system)

SL NO	Category	Description
1	Application Server in HA (2 Servers)	42 vCPU (1:1), 32 GB RAM, 500 GB Disk for OS 2 TB Disk for Data
2	Database Server in HA clustering	42 vCPU (1:1), 64 GB RAM, 500 GB Disk for OS 5 TB Disk for Data

Deployment Architecture



Proposed – Deployment Architecture

We are proposing a highly available (HA) and multi-level redundant server architecture to ensure optimal performance, reliability, and scalability. Our solution involves deploying the application on two identical, high-configuration servers, which will operate in a load-balanced manner. All incoming web traffic will be intelligently distributed between these two application servers, ensuring seamless operation and mitigating the risk of downtime in case of a server failure.

These application servers will be connected to a centralized Master Database (DB), which will handle all transactional operations, real-time queries, and critical data inputs. To enhance data redundancy and ensure data integrity, a Slave Database will be maintained, which will receive periodic synchronization updates from the Master Database.

In this architecture, all high-frequency transactions, user inputs, and frequently accessed queries will be processed directly by the Master Database to ensure minimal latency and optimal performance. Meanwhile, Management Information System (MIS) reports and other analytical processes will be executed on the Slave Database. This approach not only reduces the load on the Master Database but also enhances overall system efficiency, preventing any potential performance degradation due to resource-intensive reporting tasks.

By implementing this multi-tiered, fault-tolerant architecture, we ensure a robust, scalable, and high-performance environment that supports continuous operations while maintaining data consistency and security.

Design & Implementation Constraint

The scope of the module / functional flow might be changed during different stages of the development and implementation. This may create delay in delivery. Freezing of SRS and work flow need urgent attention. Repeated change in UI/UX may take more delay in delivery.

As the application will be bi-lingual sometimes few words exact translation may have deference. These can be corrected time to time.

User Documentation

The User Manual with detail functional activities will be provided both for system users and end users. In addition, there will be module specific user manuals to guide the user regarding how to operate in a particular module.

Dependencies

The application is a bilingual application. Therefore, the sentence of Hindi and English needs to present the same meaning. As there are some typical Hindi words are used in template so its relevant English word needs to place to provide the meaningful sentence. However, we will use native translation while development.

As specified the application will be developed & hosted in client location after one year of developed, it is highly required that requisite infrastructure is available at client location during hosting of the application with remote access of VPN to limited users.

The application will be used at the block level. So minimum 2 Mbps internet connectivity is required for accessing the application. The server will be within a good firewall system therefore the external attack on the application will be very minimal.

Developer has Dependencies for app hosting in play store approval. The delay caused due to app approval by google may delay in launch of mobile application for public. However, APK can be directly accessed by end-user.

In case of any issue, the connectivity of application and database server need to be provided to the development team for resolving the issues.

Not in Scope

The following items are considered out of scope for this project. While the application interface will support bilingual display, data entry and system messages will not be available in a bilingual format. Additionally, the development of an offline version of the application is not included. Any repeated changes or additional features beyond the scope defined in the RFP can be developed only through a separate change request, subject to mutual agreement and feasibility. Furthermore, all legacy data from the current system must be provided at the time of SRS acceptance to avoid delays in development and service rollout. The project will not be responsible for any business loss resulting from the migration of legacy data. The procurement of APIs, hardware, or demo-related infrastructure is not part of the System Integrator's (SI) scope and will be facilitated by the Board. Any additional software licensing required for the service rollout will be managed solely by the Board.

Functional Requirements

Users/Actors vs Use-Cases

The application will be used by Block, District and State offices of the board. The approval process will be from block->District->State offices. The schemes will be configured as per the criteria. The payment will be provided to the registered Worker as per the amount configured as per scheme. Including the mobile application, the website will be developed for the board .

Module 1: Construction workers / labours (In Mobile & Web app)

-  Will apply for registration.
-  Will apply for schemes.
-  Will renew on expiry of registration.
-  Will track their application.

-  Will receive message on disposal of their request.
-  Will view all sanctions
-  Manage Bio-Authentication

Worker Registration Information Will Contain Below Details:

1) WORKER_GENERAL INFO

- a) SCHEMA: Applicant name, Applicant Date, Applicant contact number, Date of birth , District, Area (Urban/Rural),Municipal corporation, Ward no., Permanent address, Present address, Father/Husband name, Adhaar number, Nature of Work, Image, CSC status, Declaration.
- b) REMARKS: Applicant name , Mobile number, Date of Birth is fetched from Adhaar API and cannot be changed. Applicant registration number will be generated automatically based on above input like District, Block ,Ward number, nature of work and date of application

2) WORKER_OTHER INFO

- a) SCHEMA: caste category (Gen/EBC/BC/SC/ST), Alternate mobile number, Adhaar card photo, ESI no. , Gender (Male/Female), Education Qualification, Marital status, Age.
- b) REMARKS: Age will be calculated based on Adhaar linked date of birth and cannot be changed.

3) WORKER_PREVIOUS EMPLOYMENT DETAILS

- a) SCHEMA: Contractor license number, Employer name, Employer address, Details of Work place, Nature of Work, Details of Work(FROM), Details of Work(TO), Total period of work, Certificate Photo.
- b) REMARKS: One worker can add multiple employers he has worked with. If Contractor license number is already registered with BOCW portal details like Employer name and address will be fetched. Work place will be selected based on projects listed by Contractor. Otherwise, worker can input details manually.

4) WORKER_BANK INFO

- a) SCHEMA: IFSC code, Bank name, Bank address, Account number, Confirm Bank account number, upload passbook, other registration, Ration card status, Ration card photo.
- b) REMARKS: Based on IFSC code bank details will be fetched and cannot be manually entered.

5) WORKER_FAMILY DETAILS

- a) SCHEMA: Family member's Adhaar number, Family member Name, address, mobile number, Date of Birth, relationship.
- b) REMARKS: Multiple family member can be added for one worker. From Adhaar number details of family member can be fetched.

6) WORKER_NOMINEE DETAILS

- a) SCHEMA: Name, Adhaar, Mobile number, Relationship, age, please tick in front of those members you want to make nominee, percentage of share.
- b) REMARKS: Multiple nominees can be added and details to be fetched from Adhaar and cannot be changed.

Worker Renewal information

- A) SCHEMA: Renewal FROM date, Renewal TO date, Membership declaration, amount, applicant registration number, Declaration status, payment status, transaction ID.
- B) REMARKS: Renewal FROM date cannot be before application date, Membership amount is calculated on the basis of number of lapsed months in registration * 0.50 paise.

Features of labour/Worker module :

SL No	Module name	Sub Module Name	Description
1	Worker Module	Login	Worker login using registration number & OTP and Aadhaar with OTP. Labour will do the Aadhar Authentication and complete the application till they get the registration number.
2	Worker Module	New Registration	The worker starts by entering the Aadhaar number, and filling up details. The registration number (ULI) will be generated after LEO/ LS Approval.
3	Worker Module	E-KYC (Adhaar Authentication Status)	Registered workers will do the Bio-Authentication from time to time and the history will be maintained.

4	Worker Module	E-Shram Portal Status	It will show if the same Aadhar number is already in the E-Shram portal. Periodically this application gets synced with E-Shram data via API.
5	Worker Module	Membership Status	It will show Approval status from LS/LEO with remarks. It will show Active / Inactive. If active, it will show registration number (ULI). else show with reason of rejection with re-apply button.
6	Worker Module	Update Membership Details	Worker can update personal details except changing Adhaar no. Once change is recorded, membership status will be unverified.
7	Worker Module	Membership Renewal	Apply to renew membership After expiry. Adding employment history, and duration of membership. This is possible when the status is inactive & only twice
8	Worker Module	Pay Membership Fees	Once membership is Active, Ask for membership fees via the payment gateway.
9	Worker Module	Apply For Schemes	labour if active can apply for scheme based on eligibility. Fill details & upload document. Ask for membership for OTP & Bio authentication. Gets Scheme Application number once complete. Send same in SMS notification.
10	Worker Module	Scheme Applications	List the scheme Application number, along with approval status from LEO, LS, Final Approval
11	Worker Module	Sanctions Orders	List of Sanctioned Scheme Approved and Payments are done against the labour or beneficiary
12	Worker Module	Payments	List of all payments completed by the board against scheme No and Sanction No

13	Worker Module	Grievance Management	Raise tickets & track tickets on any problem. Sends ticket number in SMS with details
14	Worker Module	Feedback Management	Add feedback to the board from time to time. Send a thanks message in SMS once feedback shared
15	Worker Module	Download Ecard	Downloads Ecard if membership is only active.
16	Worker Module	Board Notification	See all messages from the Welfare Board
17	Worker Module	Dashboard	Showing current membership duration, membership next renewal date, Authentication status, pending activity, numbers of schemes applied, number of approved/pending/rejected schemes, number of tickets raised, tickets resolved/pending

Note:

- No Registration number will be allocated until the worker's membership application gets approved, until then the worker will get the applicant form number to track the application status.
- Workers can select the HQ predefined grievance list to raise their grievances.
- Workers will get only their eligible scheme list in the portal to apply.

List of Schemes to be displayed & document to be collected from labour as below. Scheme to be selected by labour based on eligible scheme name & category. Once selected system will show the amount & eligibility and also ask for the required document

Labour Welfare Schemes

SL No.	Scheme	Eligibility Criteria & Documents to Be Collected	Benefit
1	Grant For the Repair of The House	Three Years of Continuous Membership & Land LPC Document, Land Current Receipt, Registered Labour with Repaired House Photo	Rs 20,000/-
2	Grant For the Purchase of a Bicycle	One Year of Continuous Membership & Document Required GST Purchase Bill, Photo Along with Cycle with BOCW Board Name Plate	Rs 3,500/- Or Amt of Receipt of Purchase of Bicycle Whichever Is Minimum
3	Grant For Purchase of Tools	One Year of Continuous Membership and On Completion of Training Provided by Skill Development & Document Required Is Certificate of Skill Development Organization	Equipped With Tools of Maximum of Rs 15,000/-
4	Death Benefit	Membership Is Required & Documents Required Death Certificate/ Dependent Family Member Adhaar Card / Dependent Family Member Certificate/ Post-Mortem Report / Preliminary Report / Oath Letter for Not Getting Any Benefit from Disaster Management / Declaration from All Dependents for Acceptance / Fir	Natural Death - Rs 2,00,000/-, Accidental Death - Rs 4,00,000/- & Rs 1,00,000/- In Case of Accidental Death Due to Disaster and Grant Already Sanctioned by LS / LEO
5	Funeral Assistance	Membership Is Required & Document Required Death Certificate/ Dependent Family Member Certificate / Post-Mortem Report or Fir.	Rs 5,000/-
6	Pension	Minimum Five Years of Membership and After Completion Of 60 Yrs of Age	Rs 1,000/- Per Month
7	Disability Pension	One Year of Membership Is Required & Document Declaration from Chief Medical Officer for Disability, Govt.	Rs 1,000/- Per Month for Permanently disabled. One Time Grant of Rs 75,000/- In Case of Permanently Disablement and One

		Hospital Treatment Certificate, Treatment Medical Bill	Time Grant of Rs 50,000/- In Case of Permanent Partial Disablement
8	Family Pension	Live Spouse of Deceased Registered Boc Worker	50 % Of the Pension or Rs 100/- Whichever Is Greater
9	Maternity Benefit	One Year of Membership Is Required	Equivalent amount of Wages Of 90 Days of Minimum Wages Rate of Unskilled Workers on The Date Of Delivery Of First Children
10	Paternity Benefit	One Year of Membership of Male Construction Worker, If His Wife Is Not Registered	Rs 6,000/- Per Delivery for The First Two Delivery
11	Cash Award	One Year of Membership Is Required & Document Required Is 10 th /12 th Mark Sheet	Rs 25,000/- & Rs15,000/- & Rs 10,000/- For the Two Children of The Registered Construction Worker for Scoring 80 % And above, 70% To 79.99% And 60% To 69.99% Marks Respectively of Class 10 And 12 Conducted by Any Board in The State of Bihar
12	Financial Assistance for Education	One Year of Membership Is Required & Document Required Admission Letter, Fees Payment Letter, Adhaar with Parent & Child Name,	(1)Total Tuition Fee After Admission In IIT/ IIM /AIIMS (2)Rs 20,000/- After Admission in B. Tech or Equivalent Courses (3)Rs 10,000 After Admission in Polytechnic/Nursing Or Equivalent Diploma Courses (4)Rs 5,000/- For Studying In Government Iti Or Equivalent Courses
13	Financial Assistance for Marriage	Three Years of Continuous Membership & Document Required Is Marriage Certificate, Couple Adhaar Card, Wedding Invitation Card.	Rs 50,000/-
14	Medical Assistance	Membership Is Required	Equivalent To C.M Medical Assistance Relief Fund
15	Annual Medical Assistance Scheme	Membership Is Required	Rs 3,000/- Per Annum

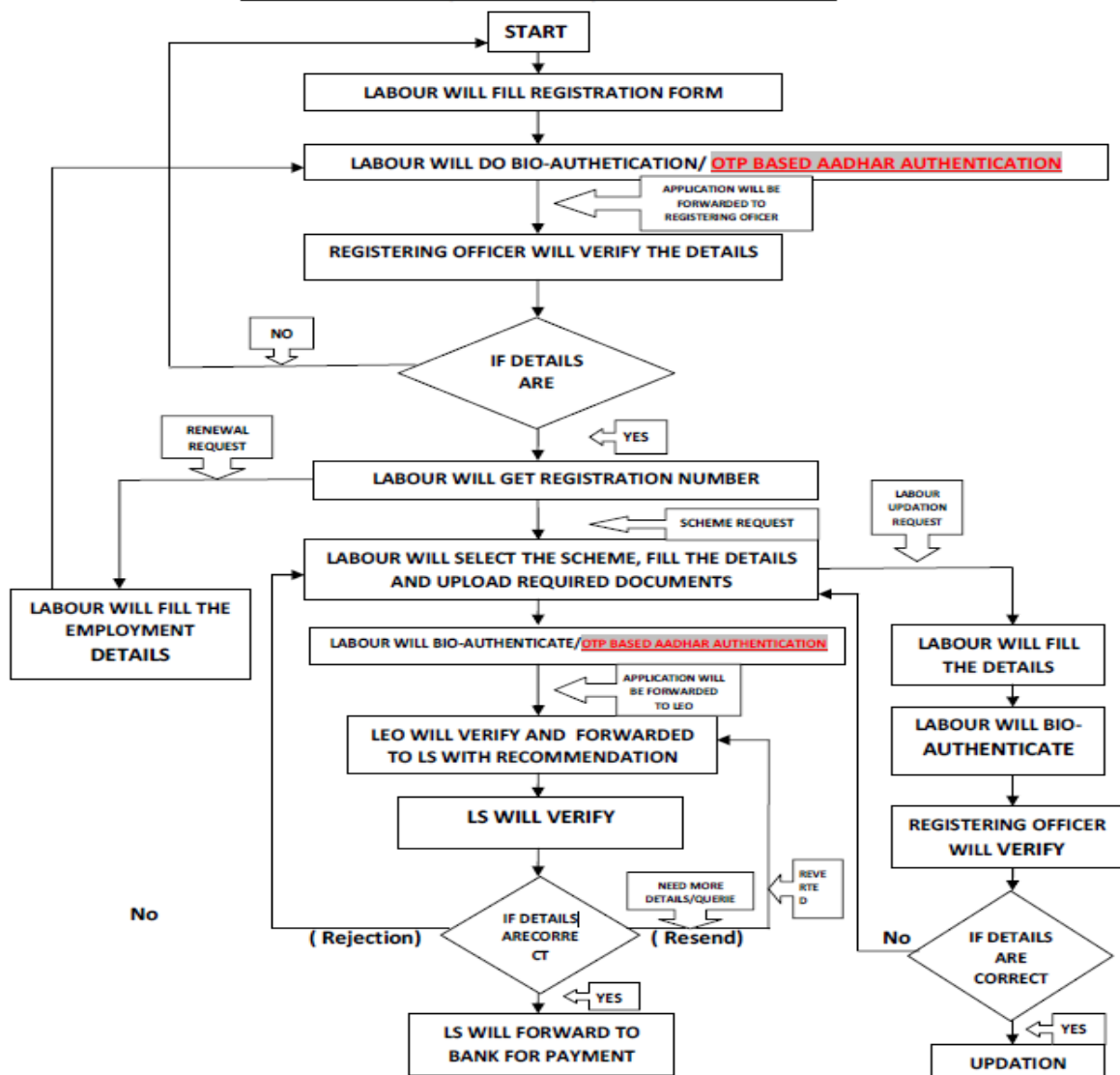
16	Annual Cloth Assistance Scheme	Membership Is Required	Rs 2500/- Per Annum
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Note:

- This application will be having 16 schemes that are approved the GoB and by this application the HQ-Admin can control the schemes limitedly.
- Apart from this, if any other schemes need to be created, then Madras Security Printers will develop and deploy the same to this application on mutual understanding.

AS PER RFP

Process Flow Diagram for Registration of Labours



Module 2: Labour Enforcement Officer (In Mobile & Web App)

- ✚ Will verify the registration application and can either approve or reject the registration application within his jurisdiction.
- ✚ Will verify the scheme application and forward it to LS with recommendations.
- ✚ Will be able to see the scheme history of workers who have applied for schemes of the very same page.
- ✚ Will approve /reject renewal application.
- ✚ Will approve/reject update requests.
- ✚ Will track applications.
- ✚ MIS Reports & Reporting
- ✚ Smart Intelligence-driven Dashboard
- ✚ View sanctions

SL No	Module Name	Sub Module Name	Description
1	Officer Login	Login	Credential or OTP-based login through mobile number
2	Officer Login	Workers - Master	View all workers under their territory with status active, inactive, verified, unverified, renewal status, and date range.
3	Officer Login	Applicants - Master	View list of fresh worker applicant under his territory, along with approval status by LEO, Approval status by LS. When LEO & LS both approve, the Applicant gets ULI.
4	Officer Login	Scheme Applications - Master	View the list of scheme Applications (under his territory) only along with their eligibility status, approval status by LEO, and approval status of LS. Also has a filter bar to separate scheme applications based on Approved, rejected, and reverted status remarks.
5	Officer Login	Renewals Master	Show all labour membership renewal application requests and the last bio-authorization/ OTP authorization done and

			the data range with the "Send Notification" Button against each member.
6	Officer Login	Workers Status Report	LEO / LS can search the worker's entire details like personal information, applied scheme details, approved schemes and E-KYC history.
7	Officer Login	Application Eligibility Status Report	Based on giving ULI, this Will list all scheme application history to take discussion of new scheme application approval. Attach Note sheet remarks by LEO against details
8	Officer Login	Reports	MIS custom reports based on his territory and assigned under RBAC.
9	Officer Login	Grievance Management	Raise tickets & track tickets on any problem. Sends ticket number in SMS with details
10	Officer Login	Board Notification	See all messages from the Welfare Board
11	Officer Login	Dashboard	Statistical data under his territory like applicant count, approved, pending, rejected, expired, renewed, schemes applied. Intelligent Dashboard.
12	Officer Login	Membership contributions – Master	List of all contributions made by the workers of the board.
13	Officer Login	CESS – Master	View all the contractors under their territory and the projects, Cess payments.
14	Officer Login	Cess – Penalty Master	List of projects by the contractor, view transactions, verify and apply penalty, also view payments.
15	Officer Login	Sanction - Master	View all sanctions under LEO territory and forward them to the LS
16	Officer Login	Helpdesk	Track and update the tickets
17	Officer Login	Feedback	Track the feedbacks

Note:

- LEO can activate and deactivate any applied schemes for the workers, however the LEO can't delete it from the system.

- Post deactivation of scheme application by the LEO, the labour can reapply for the same with valid inputs or documents.
- An LEO can be mapped with multiple territory under RBAC by HQ Admin.
- An LEO can collect and manage the cess payments against registered employer with respective projects.

Module 3: Labour Superintendent (In Mobile & Web App)

- ✚ Will verify registration application of ULBs areas and can either approve/reject the application.
- ✚ Will approve/reject/revert the scheme application forwarded to LS by LEOs.
- ✚ Will be able to see the scheme history of the workers who have applied on the very same page.
- ✚ Will forward the approved applications to bank for payment.
- ✚ MIS Reports & Reporting
- ✚ Smart Intelligence driven Dashboard
- ✚ View sanctions

SL No	Module Name	Sub Module Name	Description
1	Officer Login	Login	Credential or OTP-based login through mobile number
2	Officer Login	Workers - Master	View all workers under their territory with status active, inactive, verified, unverified, renewal status, and date range.
3	Officer Login	Applicants - Master	View list of fresh worker applicant under his territory, along with approval status forwarded by LEO.
4	Officer Login	Scheme Applications - Master	View the list of scheme Applications (under his territory) only along with their eligibility status, approval status by LS, and approval status forwarded by LEO. Also has a filter bar to separate scheme applications based on Approved, rejected, and reverted status remarks.

5	Officer Login	Renewals Master	Show all labour membership renewal application requests and the last bio-authorization/ OTP authorization done and the data range with the "Send Notification" Button against each member.
6	Officer Login	Workers Status Report	LEO / LS can search the worker's entire details like personal information, applied scheme details, approved schemes and E-KYC history.
7	Officer Login	Application Eligibility Status Report	Based on giving ULI, this Will list all scheme application history to take discussion of new scheme application approval. Attach Note sheet remarks by LEO against details
8	Officer Login	Reports	MIS custom reports based on his territory and assigned under RBAC.
9	Officer Login	Grievance Management	Raise tickets & track tickets on any problem. Sends ticket number in SMS with details
10	Officer Login	Board Notification	See all messages from the Welfare Board
11	Officer Login	Dashboard	Statistical data under his territory like applicant count, approved, pending, rejected, expired, renewed, schemes applied. Intelligent Dashboard.
12	Officer Login	Membership contributions – Master	List of all contributions made by the workers of the board.
13	Officer Login	CESS – Master	View all the contractors under their territory and the projects, Cess payments.
14	Officer Login	Cess – Penalty Master	List of projects by the contractor, view transactions, verify and apply penalty, also view payments.
15	Officer Login	Sanction - Master	View all sanctions under LEO territory and forward them to the LS
16	Officer Login	Sanction Order	list of all Sanctions Order
17	Officer Login	Sanction Order	Activate Sanctions against applied scheme
18	Officer Login	Sanction Order	Deactivate Sanctions against applied scheme

19	Officer Login	Sanction Order	Rejected sanctions against applied scheme
20	Officer Login	Sanction Payments	Generate sanction orders
21	Officer Login	Sanction Payments	Bulk payment on sanction orders
22	Officer Login	Sanction Payments	Bulk payment status
23	Officer Login	Sanction Payments	Processed Payments status
24	Officer Login	Sanction Payments	Rejected Payments status
25	Officer Login	Sanction Payments	Monthly Pension Bulk Payment posting
26	Officer Login	Sanction Payments	If any worker is paid by bank instruments like cheque or DD, its tagged to payments through sanction numbers.
27	Officer Login	Helpdesk	Track and update the tickets
28	Officer Login	Feedback	Track the feedbacks

Note:

- LS can activate and deactivate any membership or applied schemes for the workers, however the LS can't delete it from the system.
- Post deactivation of scheme application by the LS, the labour can reapply for the same with valid inputs or documents.
- An LS can be mapped with multiple territory under RBAC by HQ Admin.
- LS can process the scheme benefit payments to the sanctioned applicants.
- LS can do any deactivation of members / schemes needs an additional valid document to be attached.

Module 4: Deputy Labour Commissioner (In Mobile & Web app)

-  Will able to see and download report of their jurisdiction.
-  Will verify the details of building and cess payment done by contractor/owner.
-  MIS reports & reporting
-  Smart Intelligence driven Dashboard
-  View sanctions

SL No	Module Name	Sub Module Name	Description
	Officer Login	Login	view list of all workers with status active, inactive, verified, unverified, renewal status
	Officer Login	CESS – Master	View all the contractors under their territory and the projects, Cess payments.
	Officer Login	Cess – Penalty Master	List of projects by the contractor, view transactions, verify and apply penalty, also view payments.
	Officer Login	Reports	Custom reports as per RBAC by HQ
	Officer Login	Dashboard	all statistical data
	Officer Login	List Of Contractors	view list of contractors and manage approval.
	Officer Login	Cess Payment List	list of cess payment by contractor list
	Officer Login	Board Notification	See all message from Welfare board
	Officer Login	Penalty Management- Contractor	Manage penalty for contractors
	Officer Login	Reports	MIS Reports section
	Officer Login	Feedbacks	List all feedbacks from all stakeholders

Note:

- DLC can view MIS reports, Strategical reports.

Module 5: BBOCWW Board (HQ) (In Web App Only)

-  Will set as admin for whole BBOCWB portal
-  MIS Reports
-  Smart Intelligence driven Dashboard
-  Manage sanctions against approved scheme

SL No	Module Name	Sub Module Name	Description
1	HQ Module	Config Setting	Create grievance/ complain category & sub category
2	HQ Module	Config Setting	create & manage building types
3	HQ Module	Config Setting	create & manage districts, urban location, rural location, wards
4	HQ Module	Config Setting	Schemes, approval levels, eligibility config
5	HQ Module	Config Setting	create and manage IFSC master data
6	HQ Module	Config Setting	create and manage authority type
7	HQ Module	Config Setting	create and manage work category
8	HQ Module	Notification	Send messages to Workers (notifications)
9	HQ Module	Notification	Send Messages to Contractors (notifications)
10	HQ Module	Notification	View feedback from contractor, labour, LS, LEO, DLC, Outsiders
11	HQ Module	Notification	create & manage board directory (LEO, LS, DS, HQ)
12	HQ Module	Master Login	District Login
13	HQ Module	Master Login	Block Login
14	HQ Module	Master Login	WORKER LOGIN
15	HQ Module	Master Login	LEO LOGIN
16	HQ Module	Master Login	LS LOGIN
17	HQ Module	Worker Zone	Applicant master details
18	HQ Module	Worker Zone	Worker master details - (Filter of unverified / verified / active / inactive along with other parameters)
19	HQ Module	Worker Zone	Renewal Pending master
20	HQ Module	Worker Zone	Activate/Deactivate Worker

21	HQ Module	Worker Zone	Update Worker details
22	HQ Module	Worker Zone	Worker Payments master
23	HQ Module	Worker Zone	Manage Renewal Penalty
24	HQ Module	Worker Zone	View Worker Ecards
25	HQ Module	Contractor Zone	View Contractors / Building Owner / Employer
26	HQ Module	Contractor Zone	View Contractor wise Projects
27	HQ Module	Contractor Zone	View Cess Paid
28	HQ Module	Contractor Zone	View Cess Due
29	HQ Module	Contractor Zone	View List of Self-Declared Cess
30	HQ Module	Contractor Zone	View Penalty
31	HQ Module	Contractor Zone	View Collections
32	HQ Module	Schemes & Services	Listing Schemes & Services List. Add new Schemes with eligibility parameter & required documents for upload.
33	HQ Module	Schemes & Services	Deactivated Schemes
34	HQ Module	Sanction Order	list of all Sanctions Order
35	HQ Module	Sanction Order	Activate Sanctions against applied scheme
36	HQ Module	Sanction Order	Deactivate Sanctions against applied scheme
37	HQ Module	Sanction Order	Rejected sanctions against applied scheme
38	HQ Module	Sanction Payments	generate sanction orders
39	HQ Module	Sanction Payments	bulk payment on sanction orders
40	HQ Module	Sanction Payments	bulk payment status
41	HQ Module	Sanction Payments	Processed Payments status
42	HQ Module	Sanction Payments	Rejected Payments status
43	HQ Module	Sanction Payments	Monthly Pension Bulk Payment posting
44	HQ Module	Helpdesk/Grievance	list of Grievance Master category, sub category wise
45	HQ Module	Helpdesk/Grievance	Manage tickets and update history
46	HQ Module	Feedbacks	List all feedbacks from all stakeholders
47	HQ Module	Tenders	List all tender and add new tender option
48	HQ Module	Directory	List directory, (activate/deactivate/login)
49	HQ Module	User Management	Manage Web admin & HQ admin
50	HQ Module	News & Events	List all news. Add/activate/Deactivate news
51	HQ Module	News & Events	List all events. Add/activate/Deactivate events

52	HQ Module	Act Master	LIST all acts. Add/activate/Deactivate acts of board
53	HQ Module	Letter Master	LIST all letters. Add/activate/Deactivate letter of board (Letter can segregated as different types)
54	HQ Module	RBAC Zone	Role Creation
55	HQ Module	RBAC Zone	User Creation, Assign Role and Assign Jurisdiction Assign Mis Reports Accessibility
56	HQ Module	Dashboard	View intelligent dashboard
57	HQ Module	DMS and Media Zone	List of all media under gallery, letter , labour, contractor wise
58	HQ Module	Report Master	MIS reports like Renewal report, annual health report, scheme pendency report, senior citizen eligibility report, registration expiry report, worker 59registration status report, payment status report, legacy report
59	HQ Module	Reporting Master	View monthly List of LEO/LS reporting details like month, year, total application received, total application verified by committee, total application accepted, total forwarded to csc, total entered by CSC, total verified, accepted, done
60	HQ Module	Backup & Restoration	Option for manual data backup and restoration option

Note:

- HQ admins can create dynamic reports based multi select data type along with date range.
- HQ admin can manage website content through web admin.
- HQ admin can send notification to all stake holder by individually or group.
- HQ admin can do any deactivation of members / schemes needs an additional valid document to be attached.
- HQ admin can activate and deactivate any membership or applied schemes for the workers, however the HQ admin can't delete it from the system.

Module 6: Contractors/Building owners/Employers (In Mobile & Web App)

-  Will register themselves on portal.
-  Will register the properties whenever new construction starts.
-  Will pay the cess for construction activities.

Contractor registration include

- a) SCHEMA : Reference number, mobile number , email id, category of payers(constructor / government department /individuals/ partnership / Pvt Ltd company/ Public ltd company / proprietorship / self Help group) , Name , Address line 1 , Address line 2, Pin code , state , district , password , date of registration , government id type(CIN / PAN / ADHAAR), government id number , mobile OTP authentication Status , Email id authentication.
- b) REMARKS : CIN /PAN to be used for private limited company and rest for others Adhaar number has to be used.

Contractor Project addition

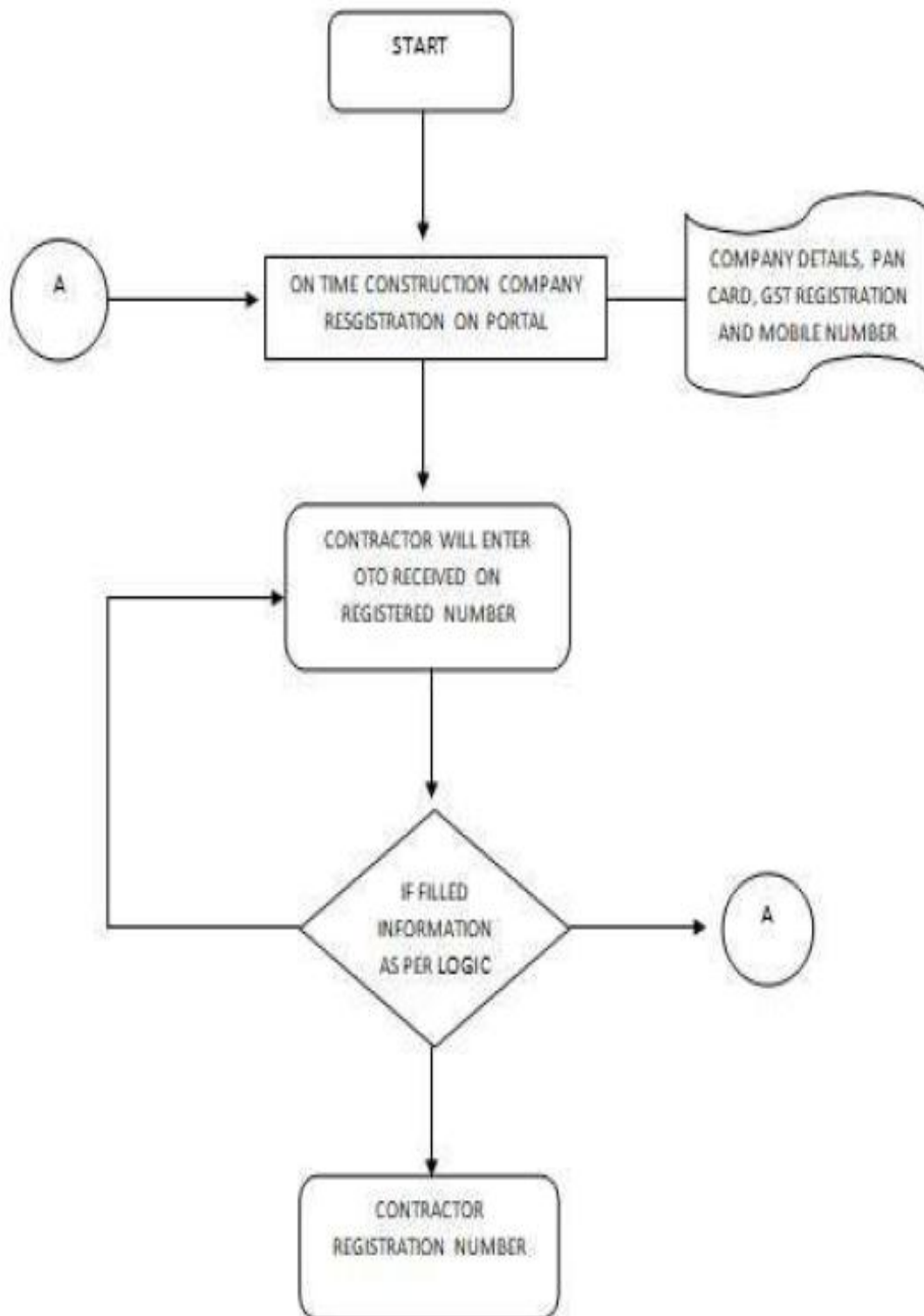
- a) SCHEMA : State(BIHAR), District ,Multi district project (YES/NO) , Block Municipal corporation , Establishment registration number as per BOCW act, Site address , Site electricity CA number , number of labours , Project description , Project start date , Actual / Estimated finish date , Period of construction , type of project , Site contact person , Mobile number , KML file.
- b) REMARKS : Constructor can add more than one project .

Contractor Cess Payment information

- a) SCHEMA : Project ID , Payment option , Mode of payment, payment details , paid amount.
- b) REMARKS : Details of view all transactions, Payment option under cess module after adding project.

As Per RFP

Process Flow Diagram for Contractor on Boarding



SL No	Module Name	Sub Module Name	Description
1	Employer Login	Contractor Registration	Employer will fill the form to get registered.
2	Employer Login	Login	Employer can login with Email OTP or Mobile OTP
3	Employer Login	Profile Master	View their profile and change the basic details. (Conditional data only)
4	Employer Login	Project Master	Add and Manage projects under their contraction.
5	Employer Login	Cess Master	View all Cess Payments against project listed with action button.
6	Employer Login	Penalty Master	View Penalty and remarks added by LEO
7	Employer Login	Dashboard	All data related to their Cess payment and projects
8	Employer Login	E-KYC	Complete the E-KYC
9	Employer Login	Board Notification	See all message from Welfare board
10	Employer Login	Helpdesk/Grievance	Employer can raise and track their grievance.
11	Employer Login	Feedbacks	Employer can share their feedback.

Module 7: Content management System (Web & Mobile app) for BBOCWW Website management

-  Manage website all content / dynamic pages
-  Manage Tenders
-  Manage letter
-  Manage documents
-  Manage tickets / grievance
-  Manage Directory
-  Will register the properties whenever new construction starts.
-  Will pay the cess for construction activities.

SL No	Module Name	Sub Module Name	Description
1	Web Admin	Page management	List Of Page in Website front page along with edit button.
			This will cover About us page, organogram, RTI, Q&A, FAQ, T&C, refund, disclaimer. Web admin can edit the content .
2	Web Admin	Directory Management	List of Employee associated with board with a search-bar.
			It will display active staff of board with Division name, district, staff type (DLC/ALC/LEO/LS), Contact Number, Email Id.
3	Web Admin	Acts Management	List of Act under BOCW, adding new acts & disabling added acts)
4	Web admin	Scheme & Services	List of Schemes, adding new acts & disabling added acts)
			This will be displaying only schemes added by admin.
5	Web Admin	Contact us Page	Update contact address of board to be displayed in app and website
6	Web admin	Banner Management	adding banner for home page
7	Web Admin	News & Events	List of News & events
8	Web admin	Letter Management	List of Letters
9	Web Admin	Forms Management	List of forms for end users
10	Web Admin	Tender Management	List of tenders under BOCW, adding new tenders
11	Web Admin	Link Management	List of Links to be display in website
12	Web Admin	Media / Gallery Management	Managing images to be added to gallery album wise.
			Create a gallery & add image

13	Web admin	Chat box Management	Enable/Disable Chat box option
14	Web Admin	Grievance management	List of tickets and manage tickets
15	Web Admin	Feedback management	List of feedback given
16	Web Admin	Change Password	Web admin can change the password
17	Web Admin	Configure dashboard	Facility to configure dashboard with A for smart use
18	Web Admin	Director Message	Configure director message, other top management
19	Web Admin	Login Page	Web admin will login with captcha and OTP will be got to Email/Mobile number-based login to registered number)

Web Portal Options

SL No	Module Name	Sub Module Name	Description
1	Web portal/Mobile App	Home	Display header, banner, page menu (about us, organogram, schemes & services, act, directory, gallery, Downloads, RTI, FAQ, Contact us),
2	Web portal/Mobile App	Home	Display footer information like important links, FAQ, Disclaimer, policy, contact us, TERMS & CONDITIONS, Refund & Cancellation Policy
3	Web portal/Mobile App	Home	Display director detail and Message added by web admin
4	Web portal/Mobile App	Home	Display icon of Chabot
5	Web portal/Mobile App	Home	Display icon of Grievance button, feedback Button

6	Web portal/Mobile App	Home	Display icon of language change
7	Web portal/Mobile App	Home	Display Tab for recent active tenders with read more button
8	Web portal/Mobile App	Home	Display Tab for recent active News & Events with read more button
9	Web portal/Mobile App	Home	Display Tab for recent active Letter with read more button
10	Web portal/Mobile App	Home	Highlighting Button like Register Labour, apply for scheme, Track Applications, Download Ecard, labour login, officer login, Pay Cess, CSC Login, Contractor Login
11	Web portal/Mobile App	About Us	Display details of about us set by web admin
12	Web portal/Mobile App	Organogram	Display details of organogram set by web admin
13	Web portal/Mobile App	Schemes	Display details of scheme set by web admin
14	Web portal/Mobile App	Act	Display list of acts set by web admin
15	Web portal/Mobile App	Directory	Display list of staff in labour department with simple search filter , specifying division, district, staff type, name, email, phone
16	Web portal/Mobile App	Photo Gallery	Display album and click on album will show photos
17	Web portal/Mobile App	RTI	Display details of RTI information entered by web admin
18	Web portal/Mobile App	FAQ	Display details of FAQ information entered by web admin
19	Web portal/Mobile App	Contact Us	Display details of Contact Details along with map entered by web admin

20	Web portal/Mobile App	Register Labour	Display list of all labour registered as per district, area, municipal corporation, ward number.
21	Web portal/Mobile App	Apply For Scheme	Labour can provide registration number, authenticate and view schemes he is eligible.
22	Web portal/Mobile App	Track Application	Labour can provide registration number, authenticate and see all sanctioned approved schemes
23	Web portal/Mobile App	Download Ecard	Labour can provide registration number, authenticate and download ecard
24	Web portal/Mobile App	Downloads	List of forms uploaded by web admin will be displayed
25	Web portal/Mobile App	Grievance Button	one tab to add grievances, other tab to track the grievance ticket status
26	Web portal/Mobile App	Feedback Button	To display a form to take feedback from user and display to web admin
27	Web portal/Mobile App	Language Button	To toggle the language of the website
28	Web portal/Mobile App	Chat Button	Display icon to open Chabot that will guide end user . This will be auto populated once the portal is in use for 5-6 months.
29	Web portal/Mobile App	Disclaimer	Display details of disclaimer added by web admin
30	Web portal/Mobile App	Policy	Display details of Policy added by web admin
31	Web portal/Mobile App	Officer Login Button	display login for officers
32	Web portal/Mobile App	Pay Cess Button	Display Contractor login option
33	Web portal/Mobile App	CSC Login Button	Display CSC login page
34	Web portal/Mobile App	Contractor Login	Display contractor login option

35	Web portal/Mobile App	Terms & Conditions	Display page details added by web admin for T&C
36	Web portal/Mobile App	Refund & Cancellation Policy	Display page details added by web admin for refund page

Functional Control Flow

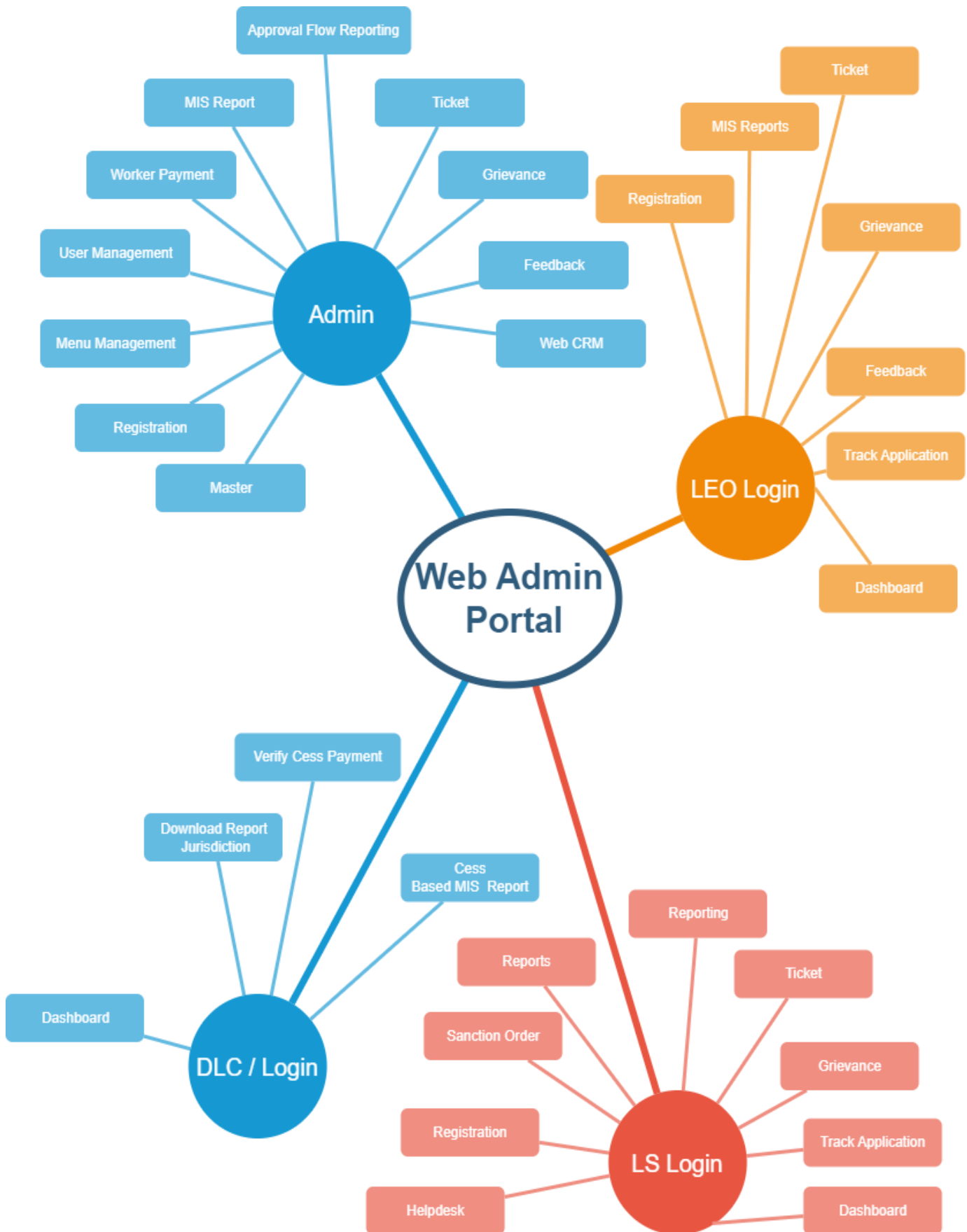
Web Portal / Mobile App



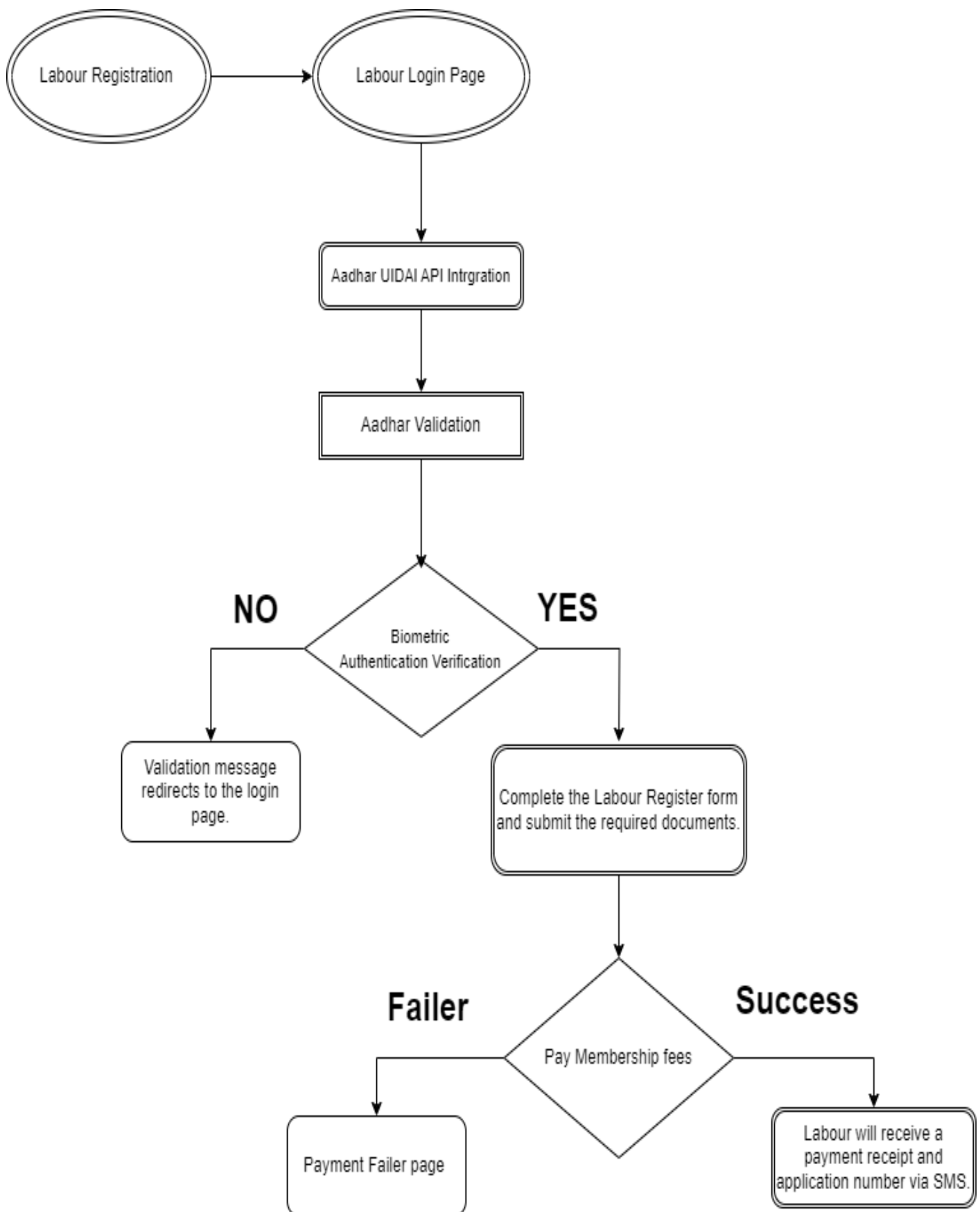
Website



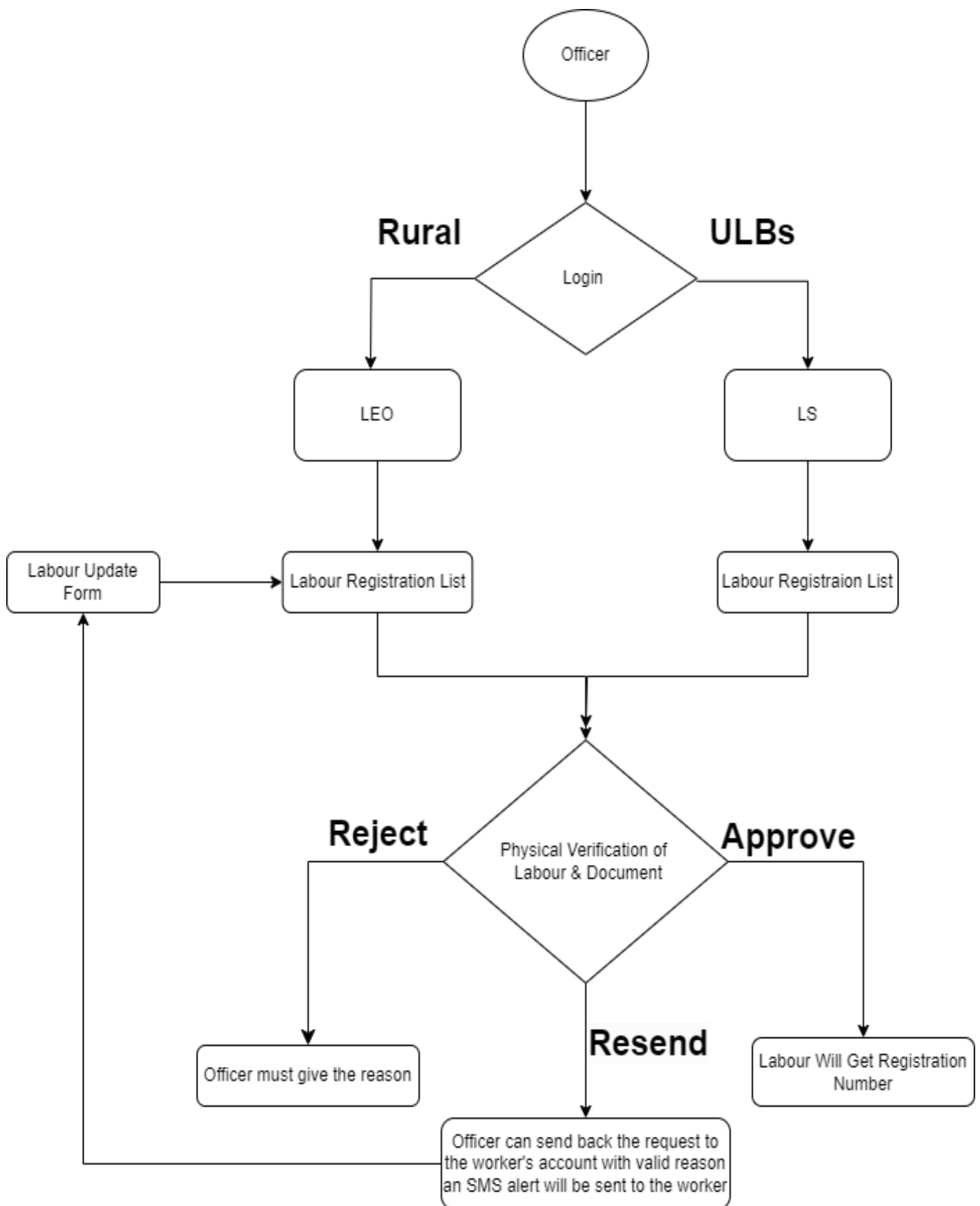
Web Admin Portal



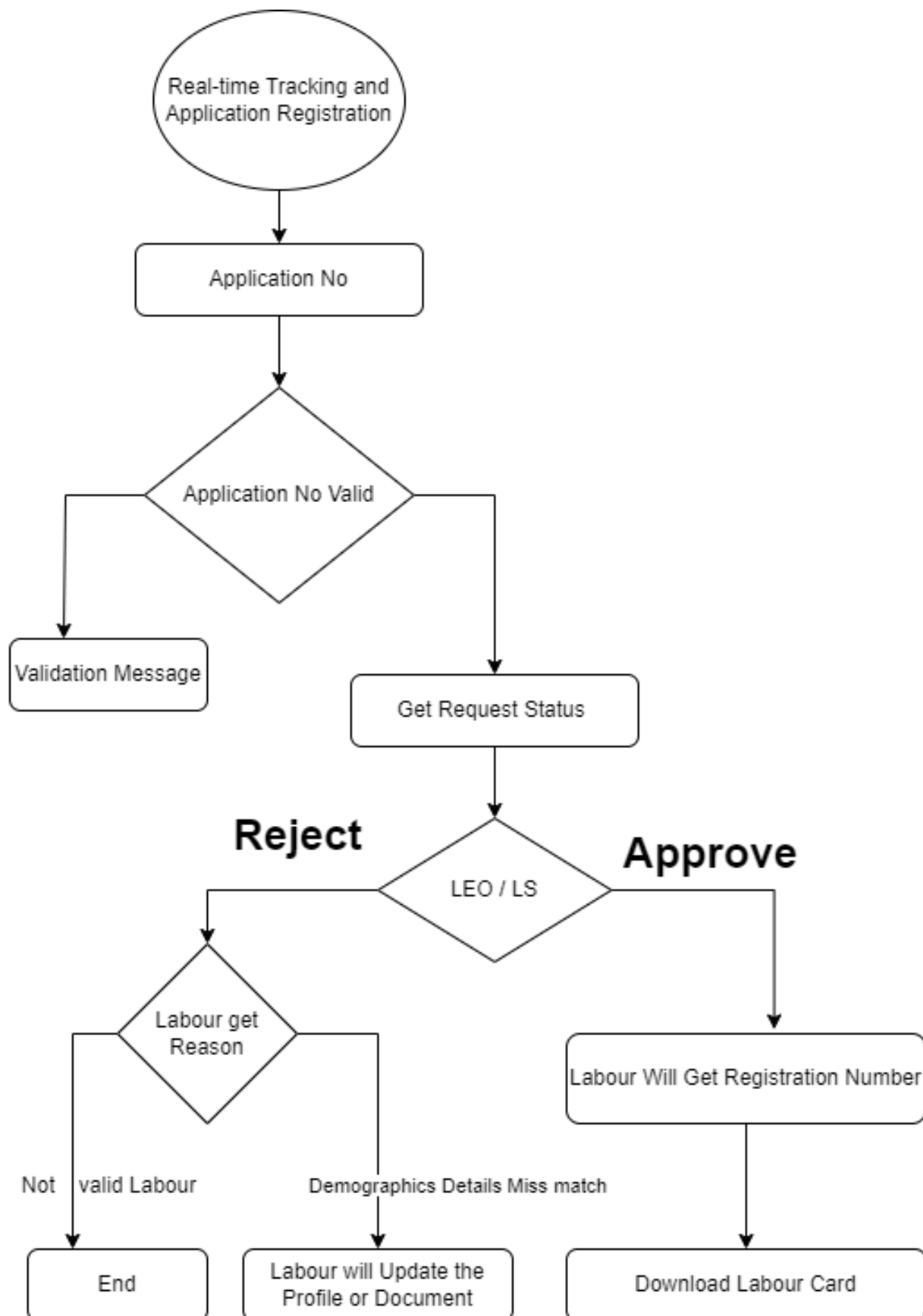
Labour Membership Registration



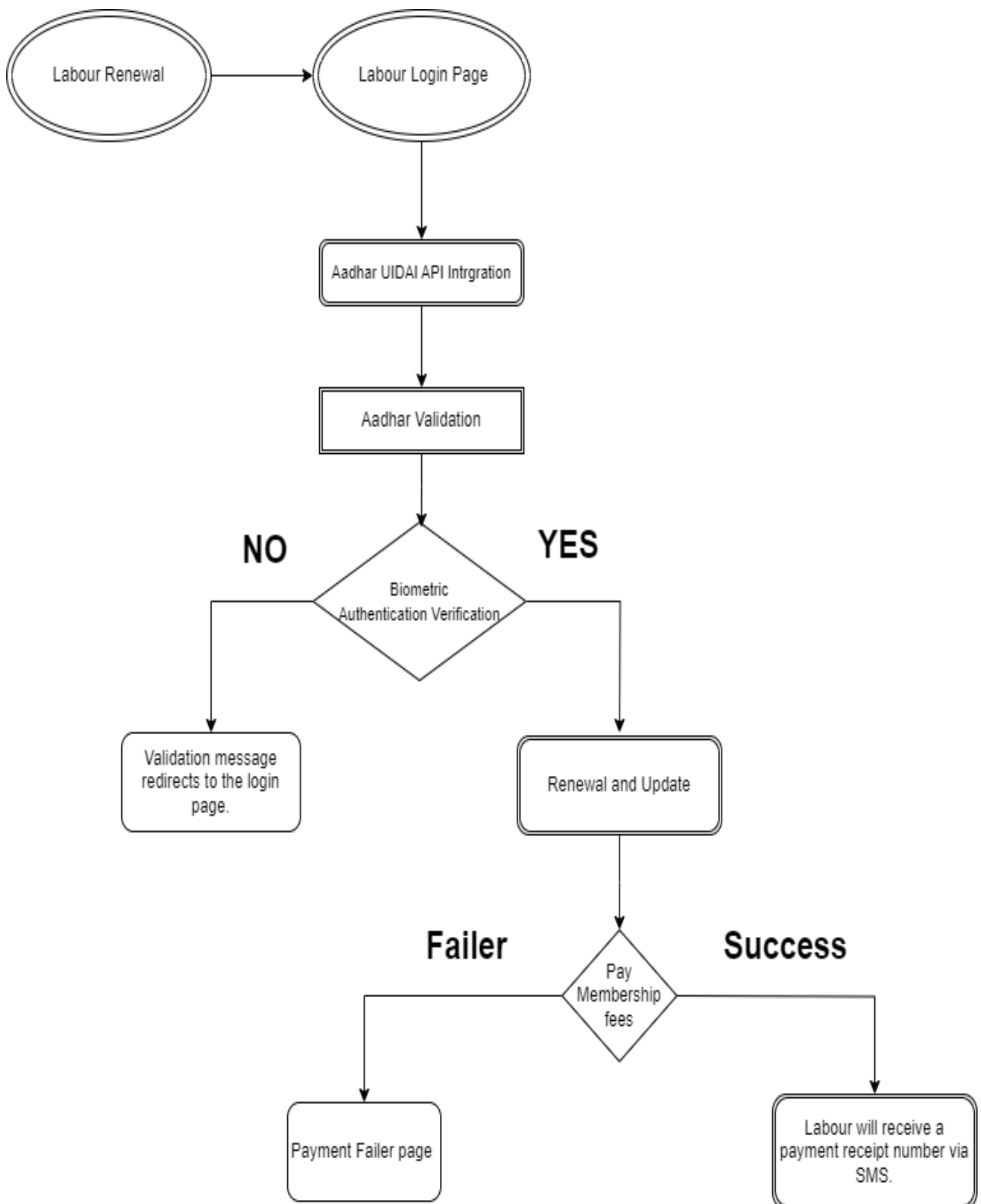
Labour Membership – Approval



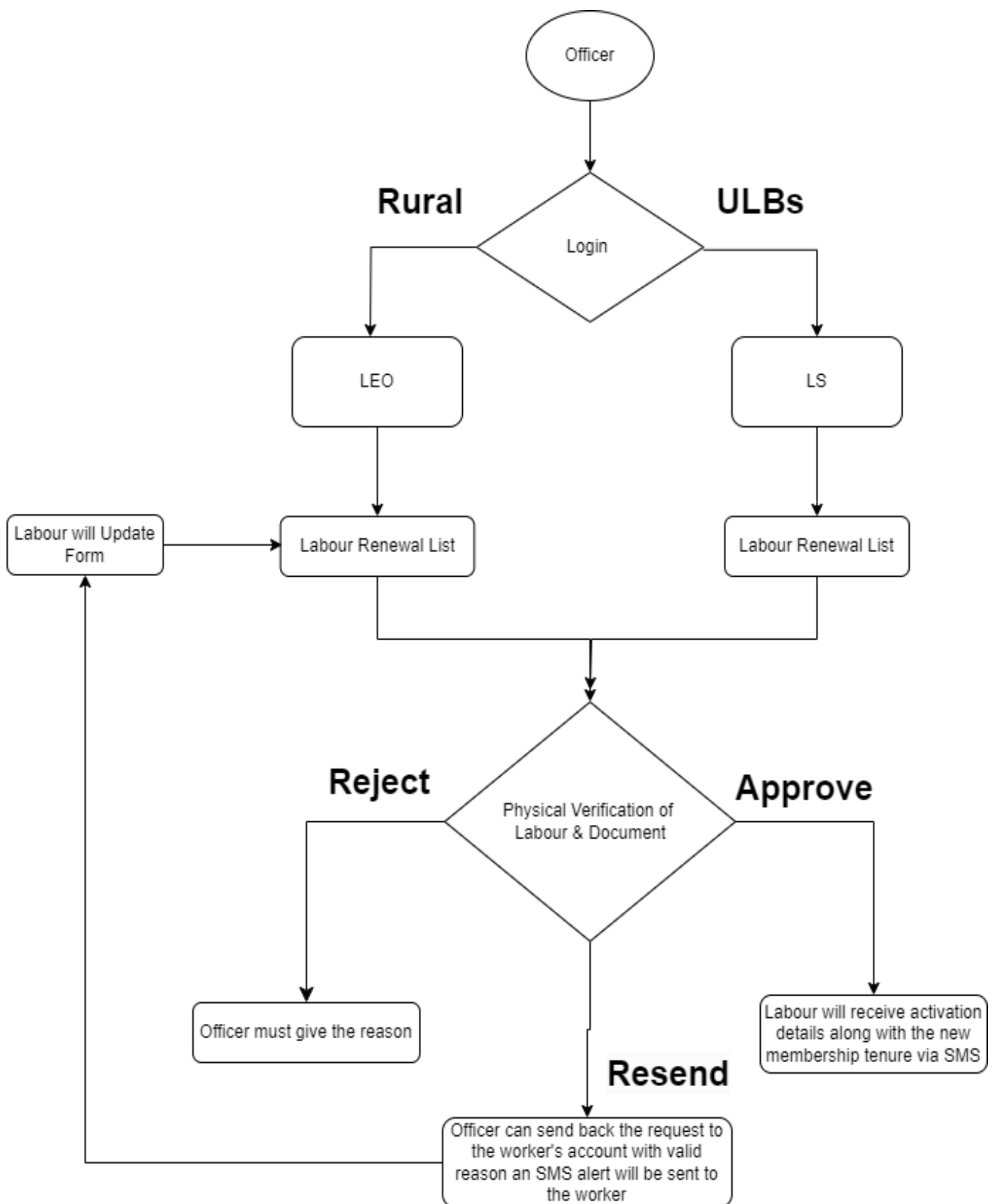
Labour Membership Registration – Real-Time Tracking



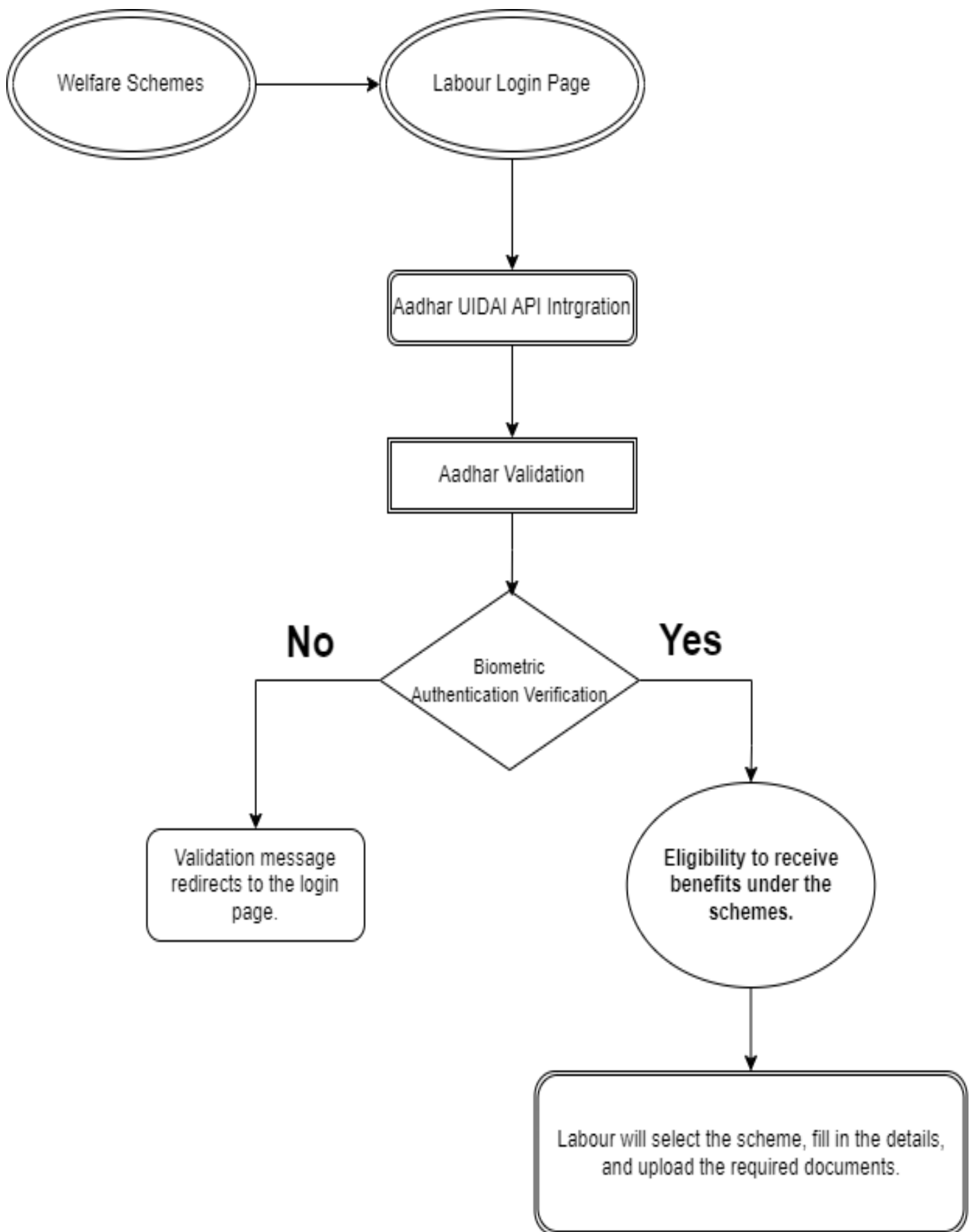
Labour Membership Renewal



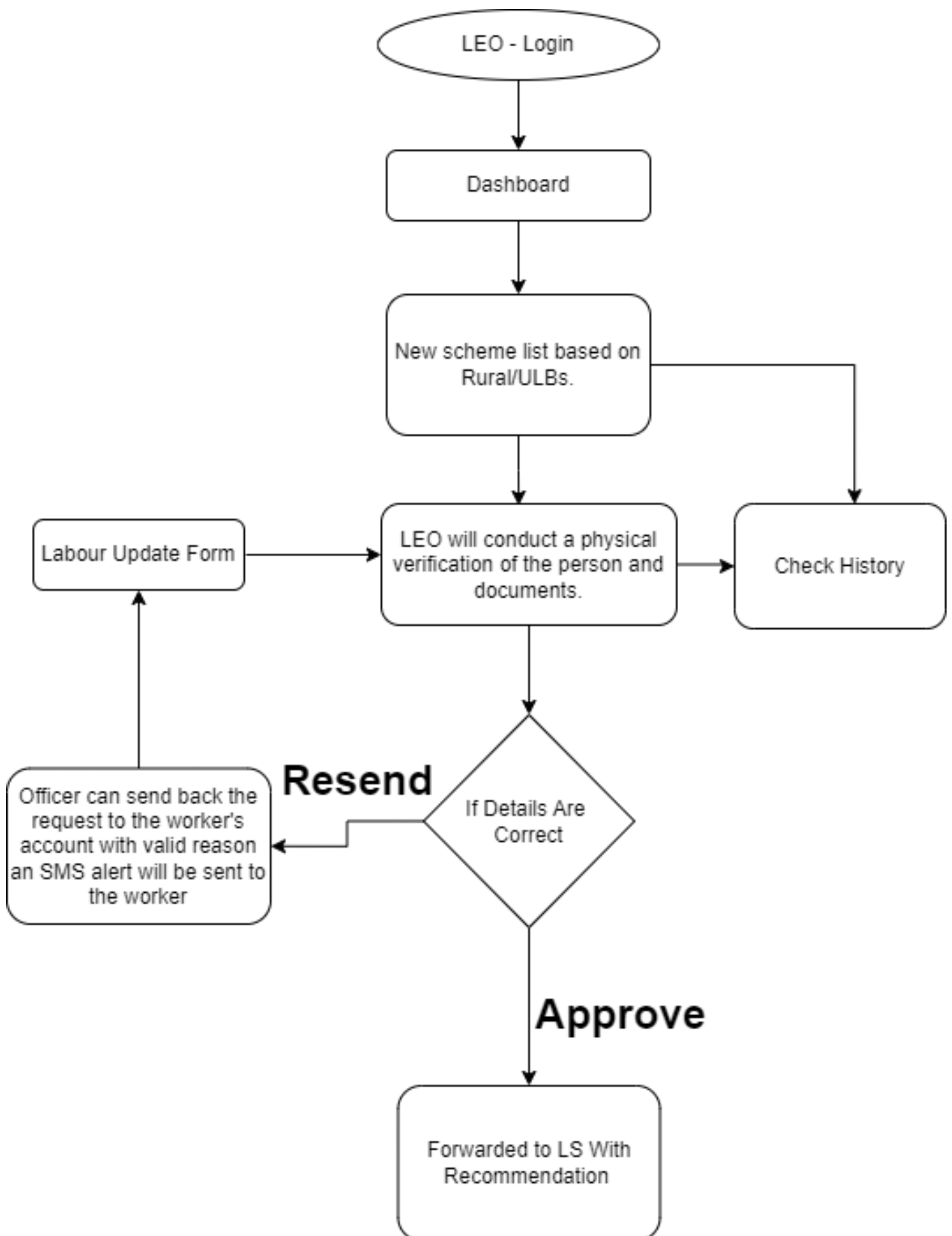
Labour Membership Renewal Approval



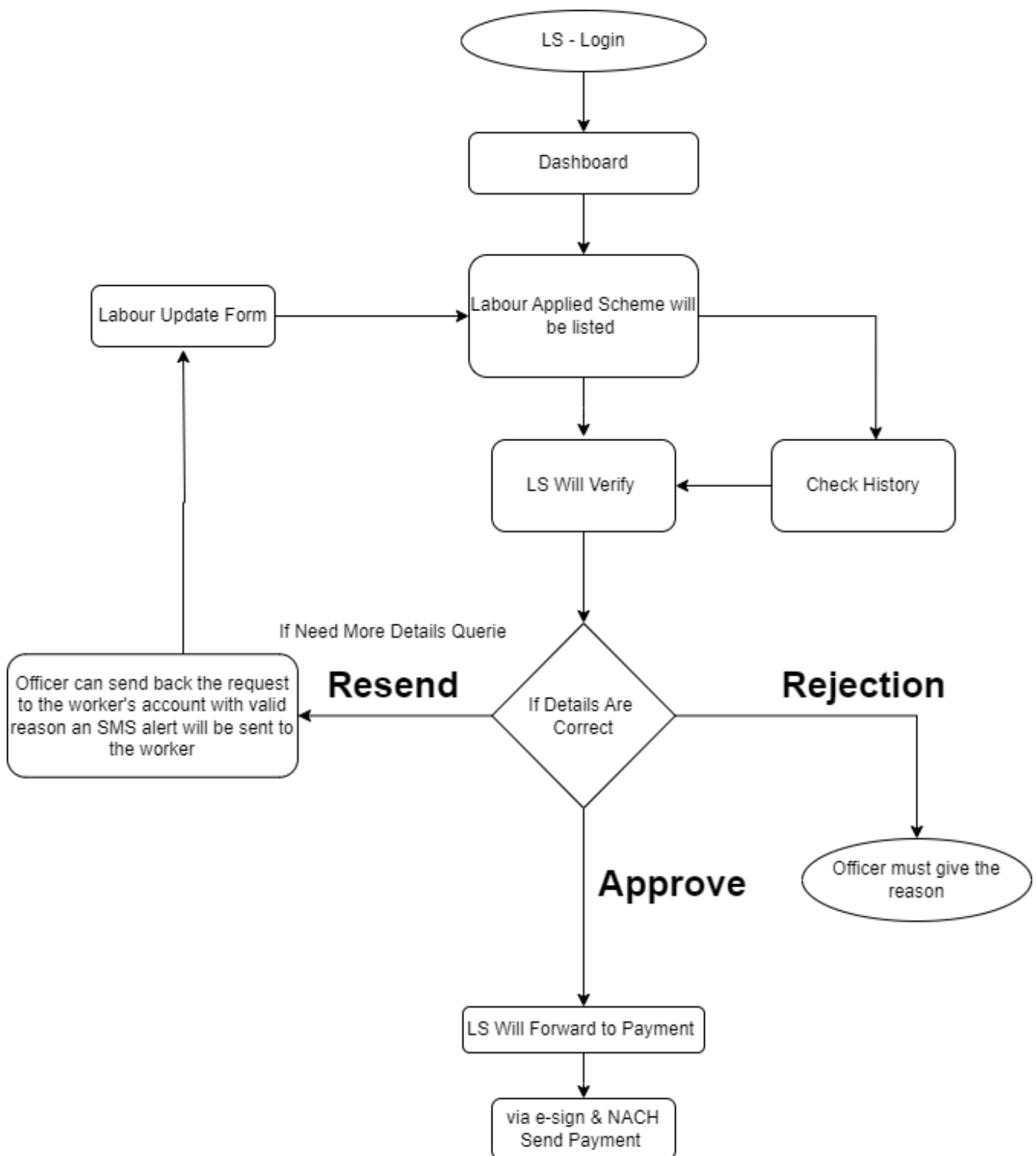
Labour Applies for Scheme



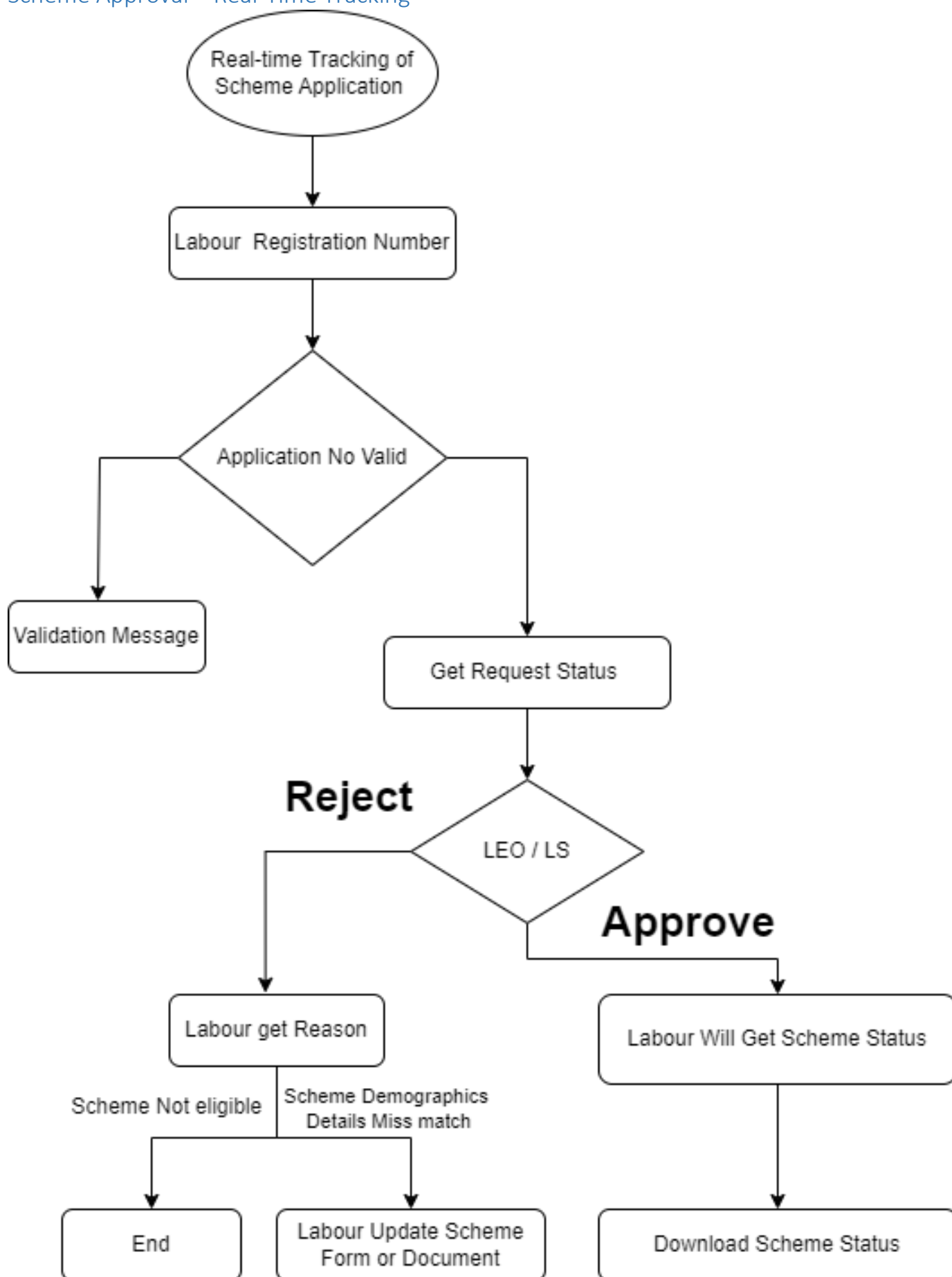
LEO Forward



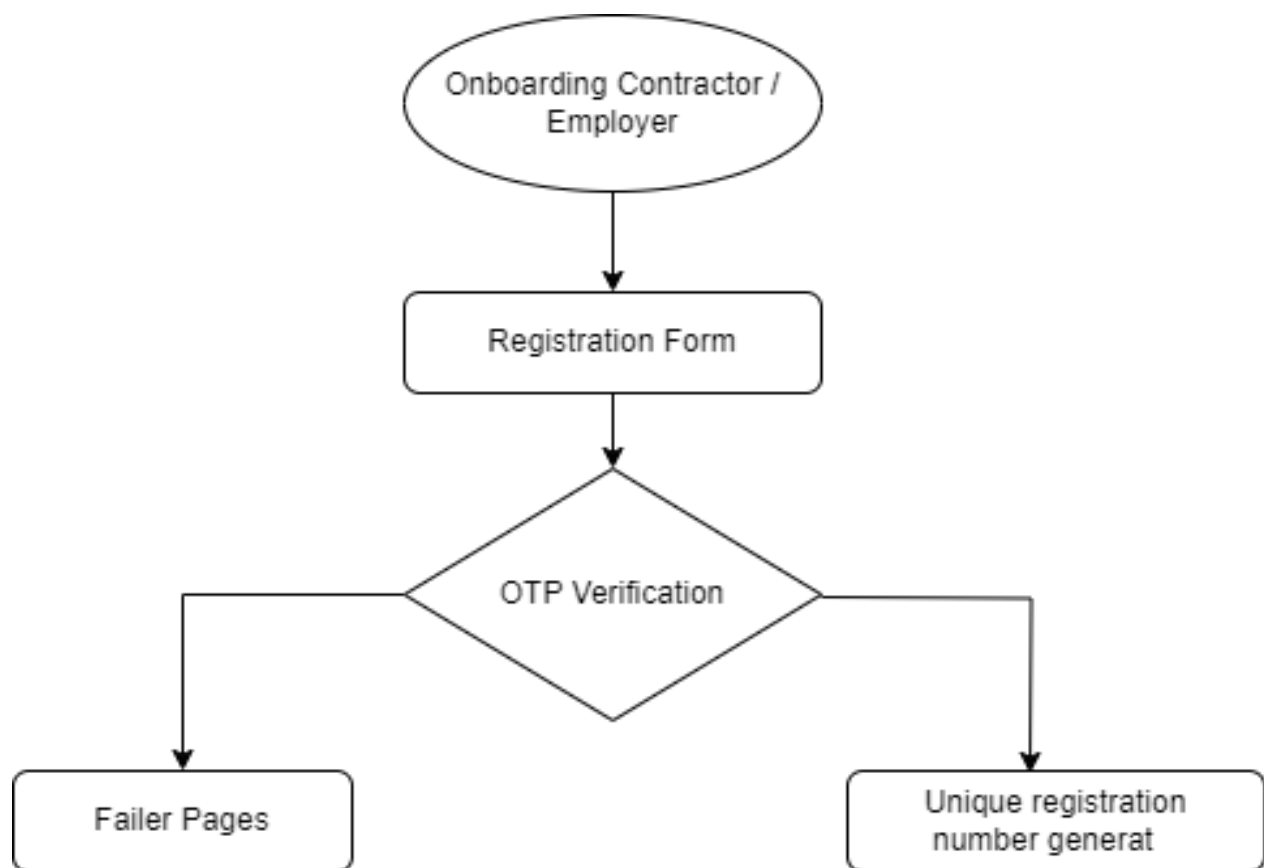
LS Approval



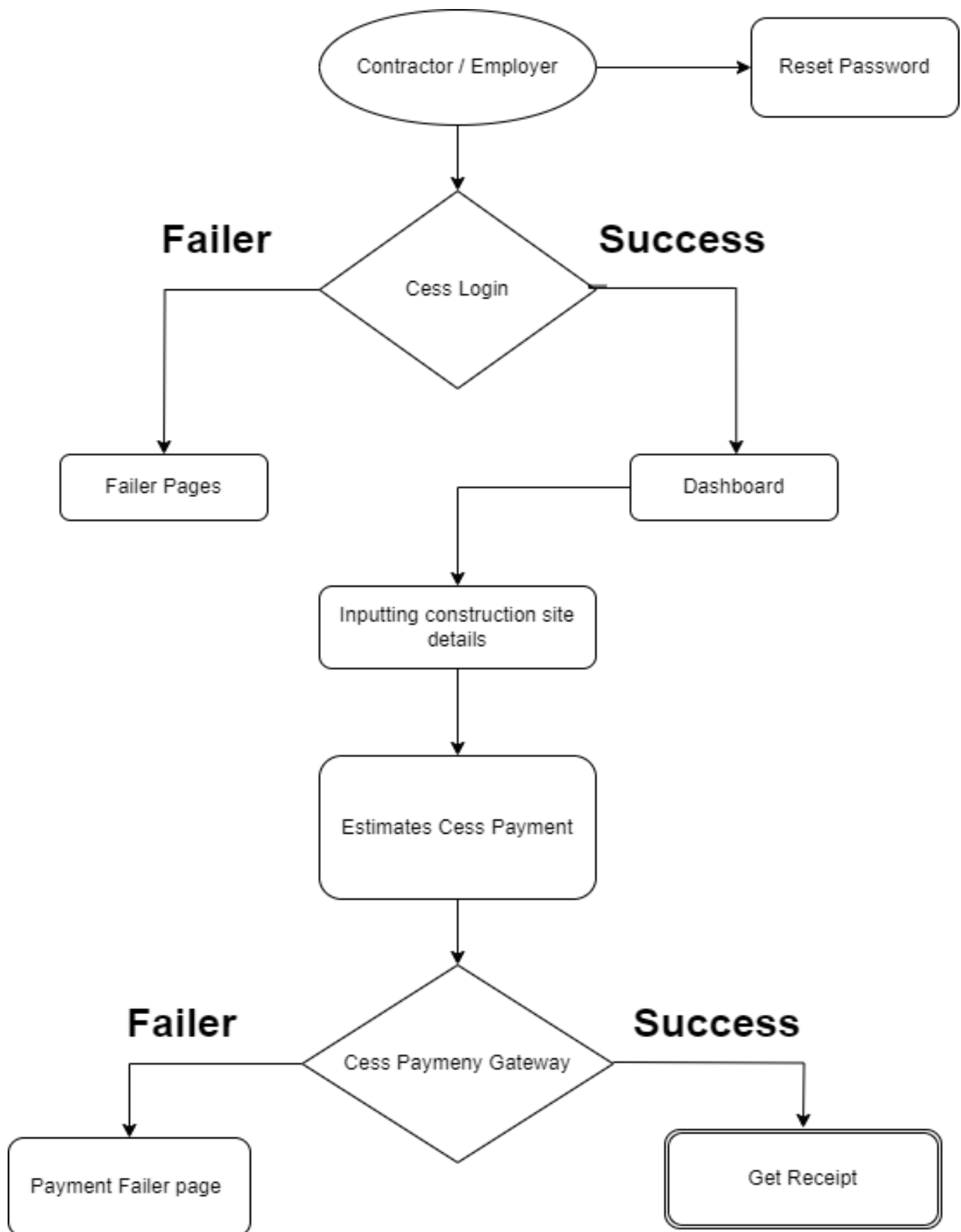
Scheme Approval – Real-Time Tracking



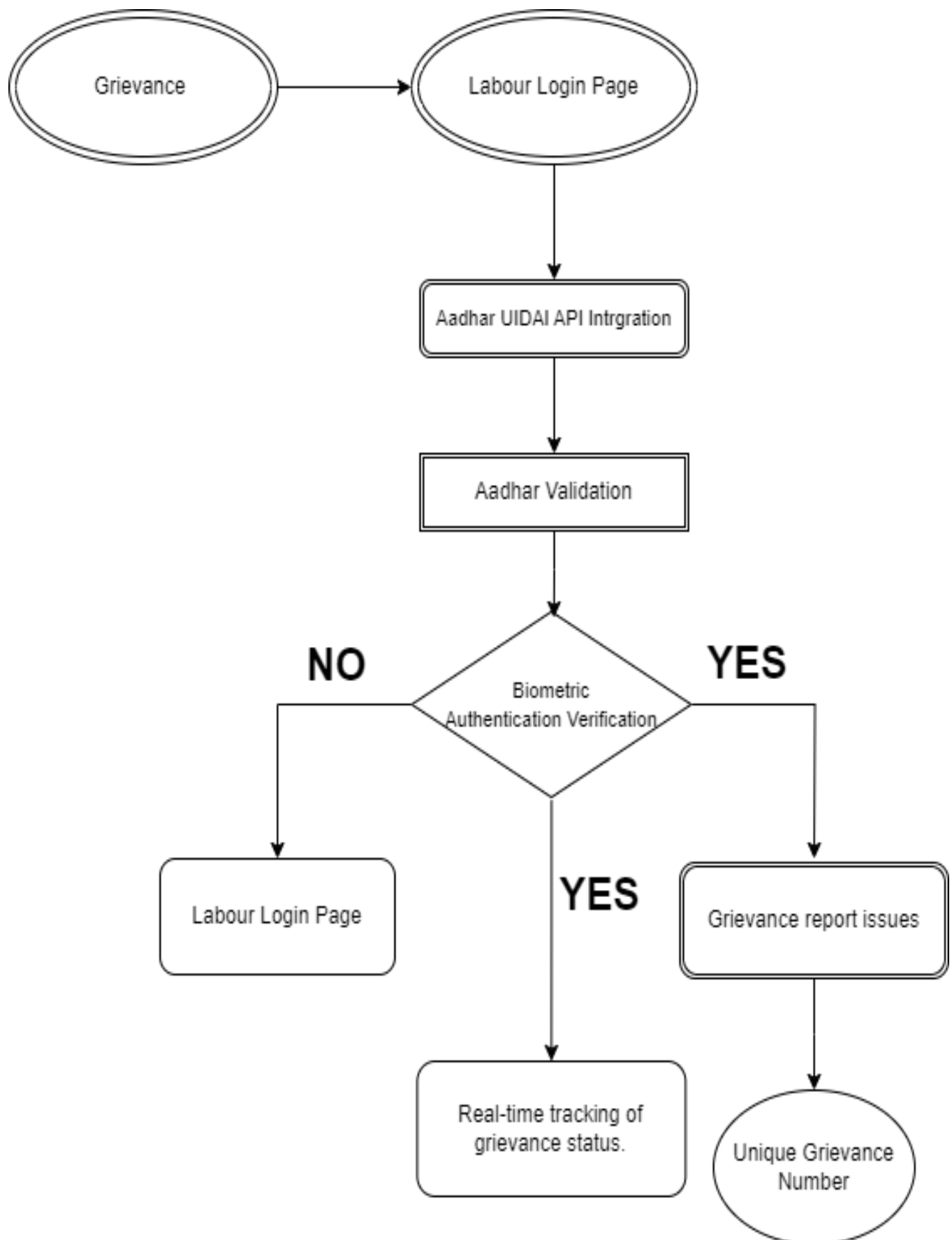
Employer Registration



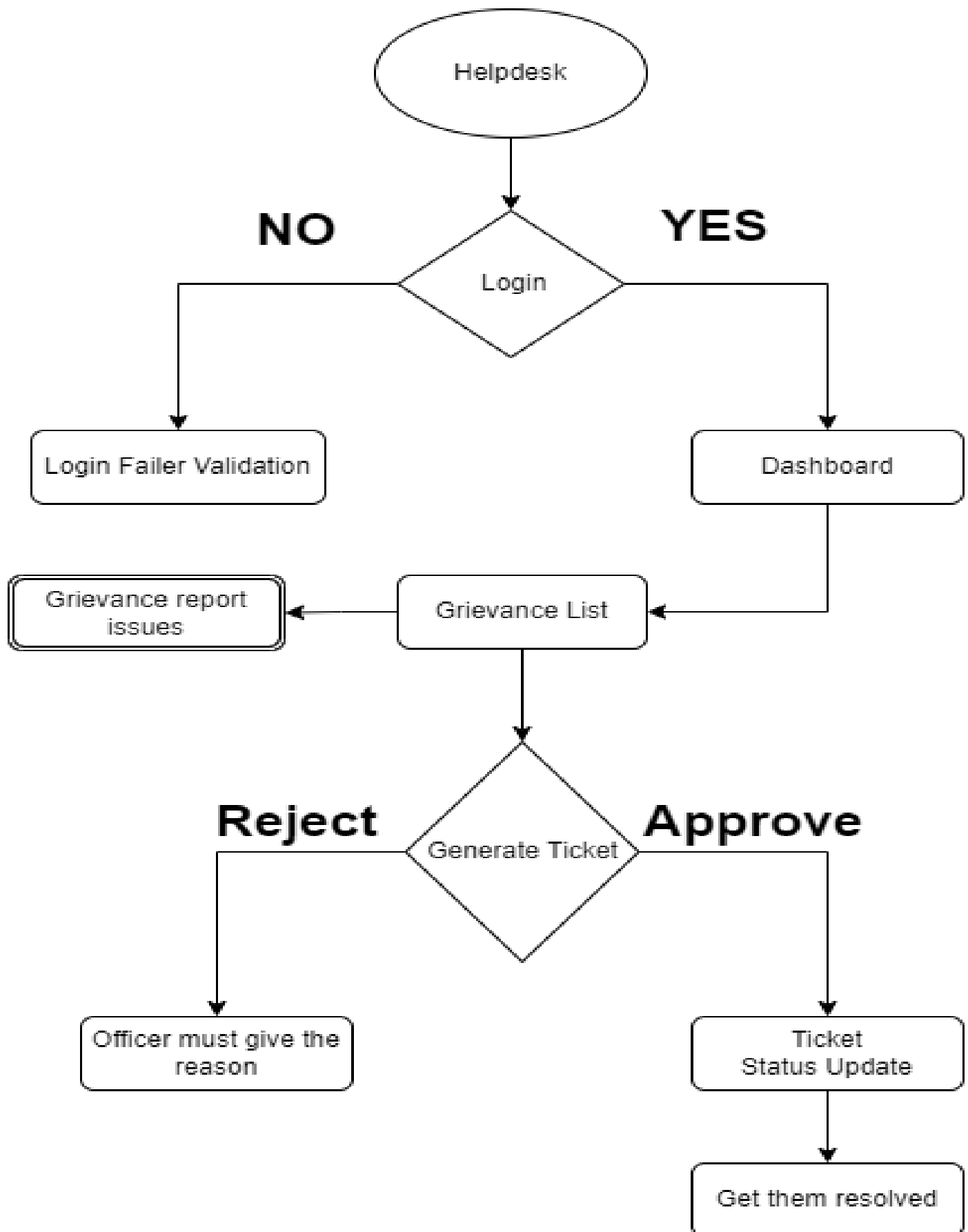
CESS Payment



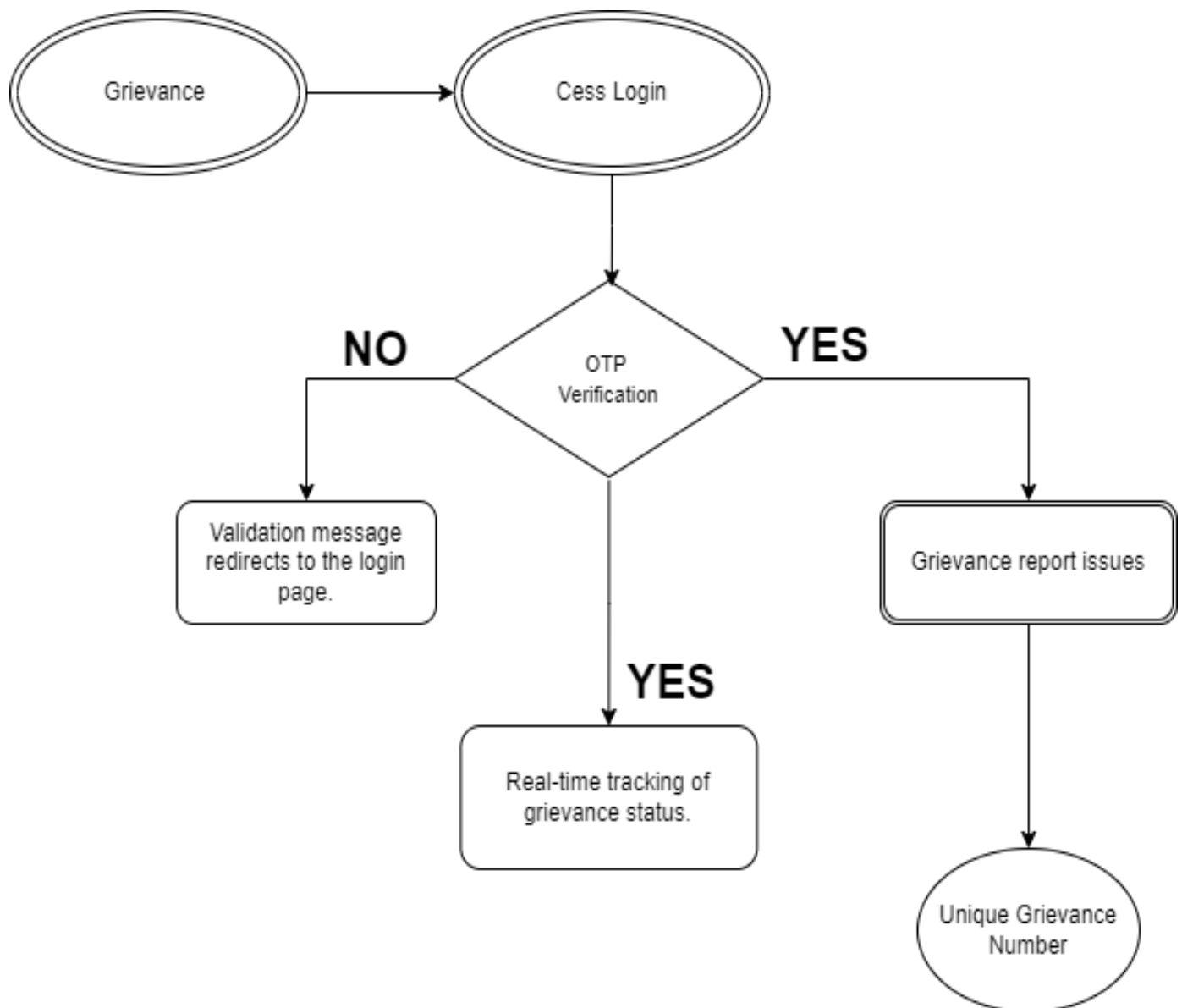
Labour Grievance



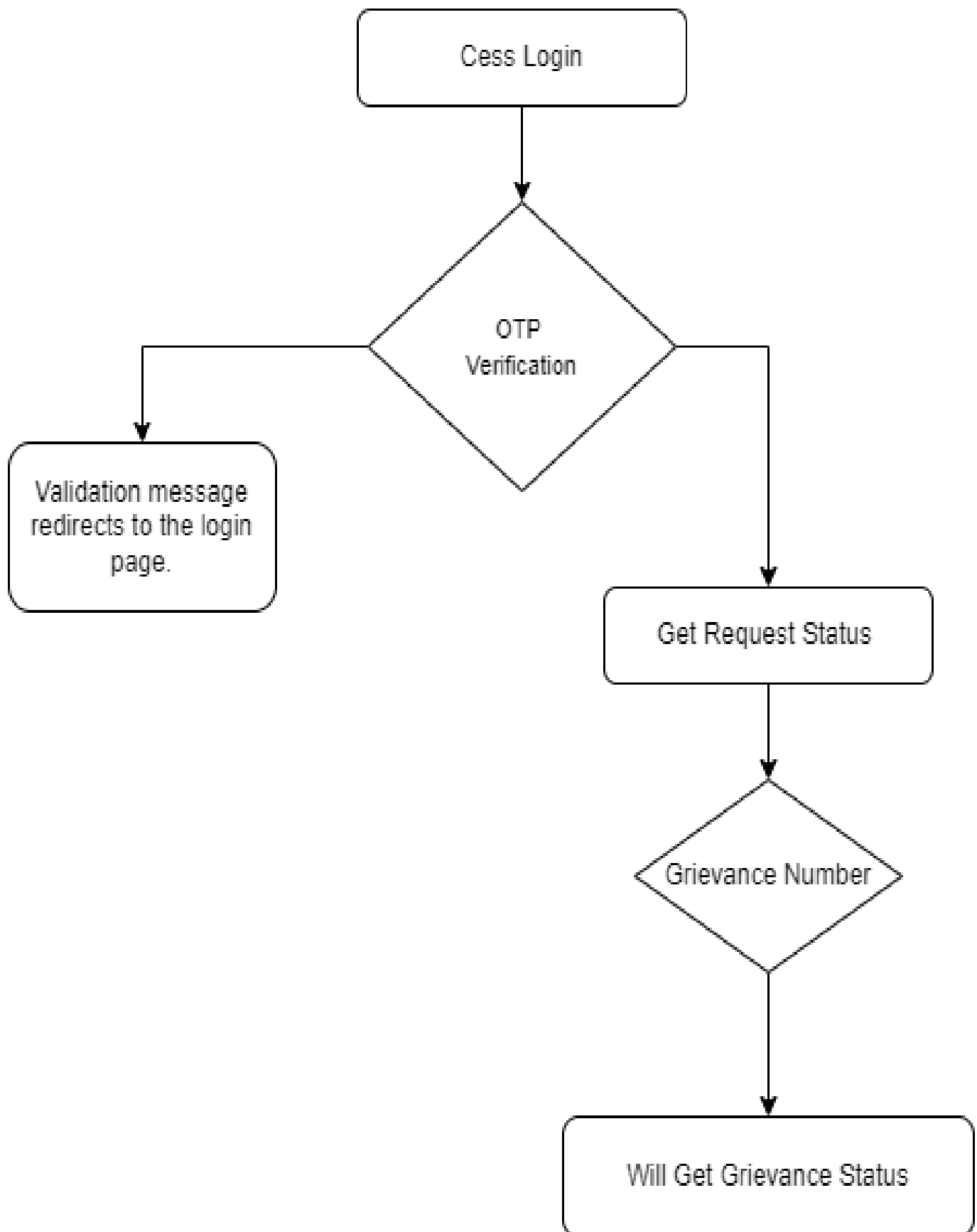
Ticket Flow (Grievance)



Employer Grievance

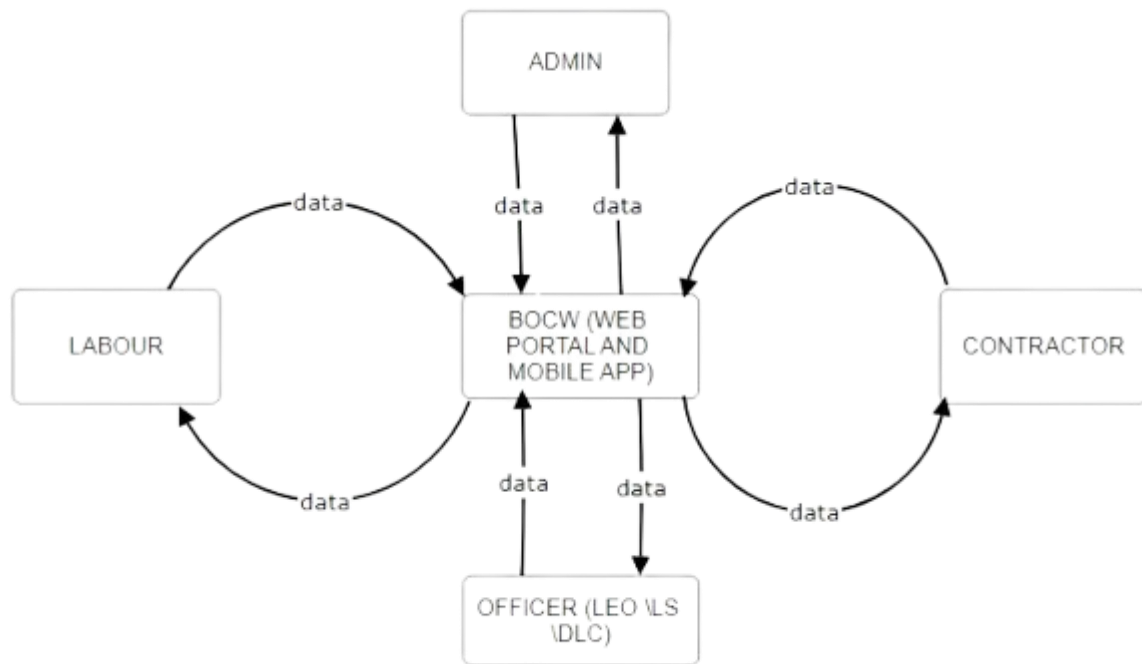


Cess Payment – Real – Time Tracking

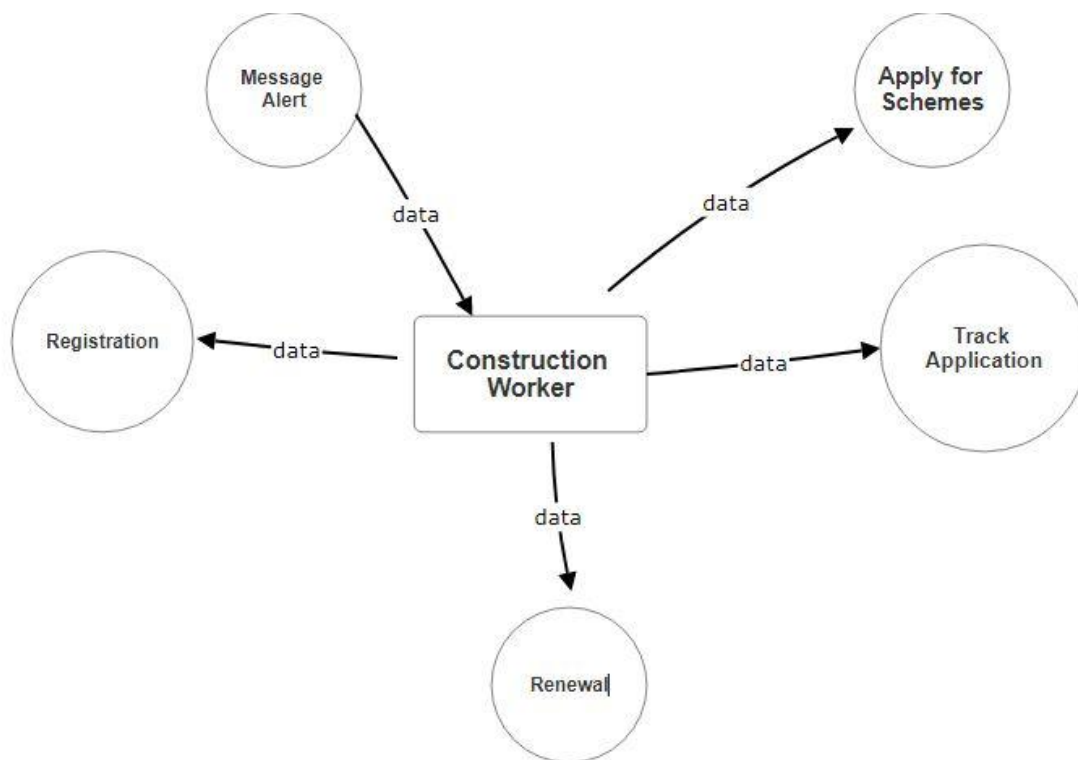


Data Flow Diagram

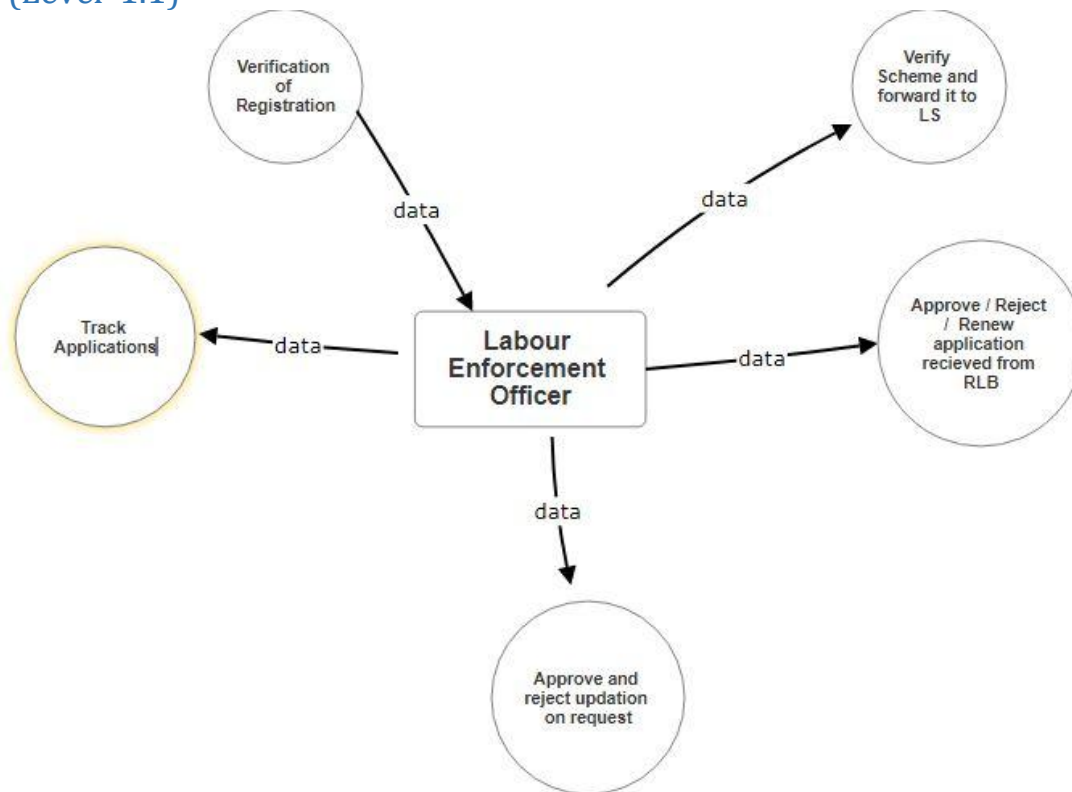
Context Diagram (Level 0)



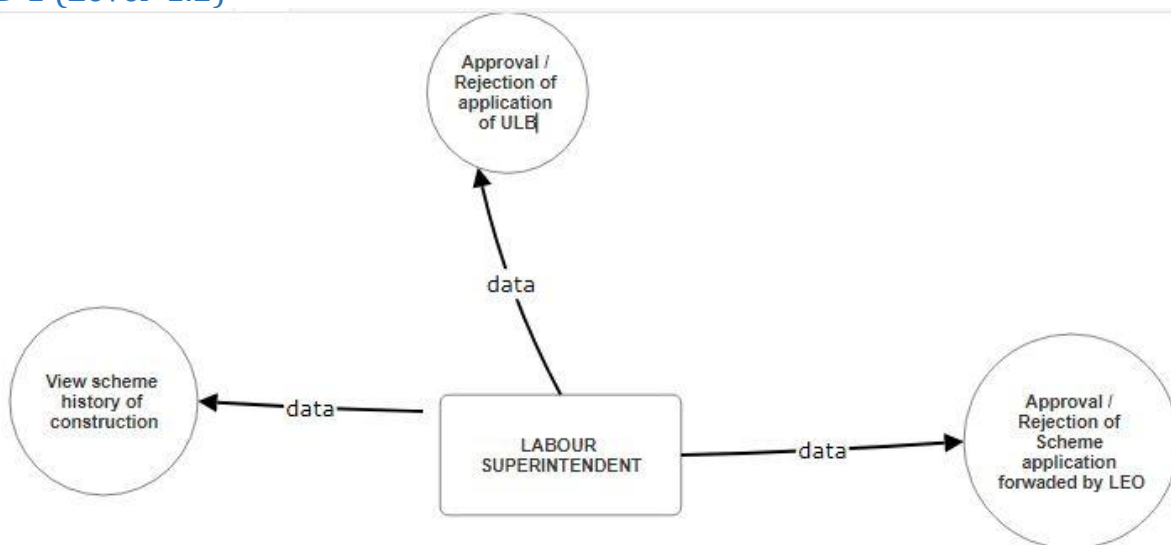
DFD 1 (Level 1)



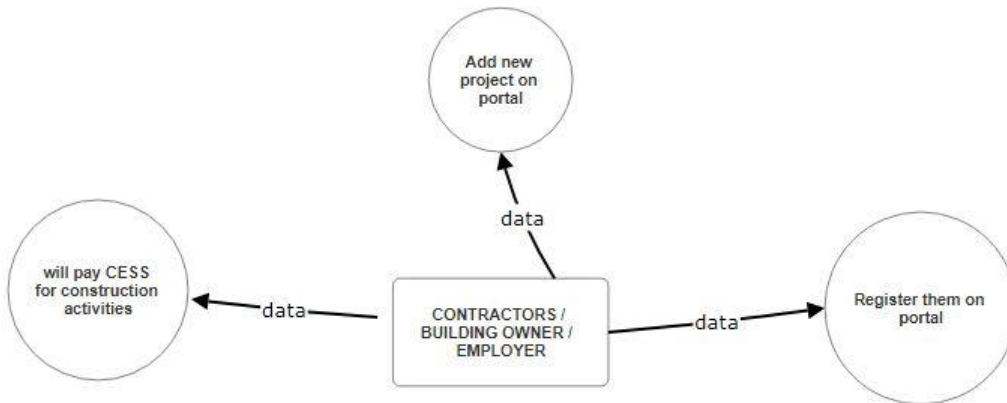
DFD 1 (Level 1.1)



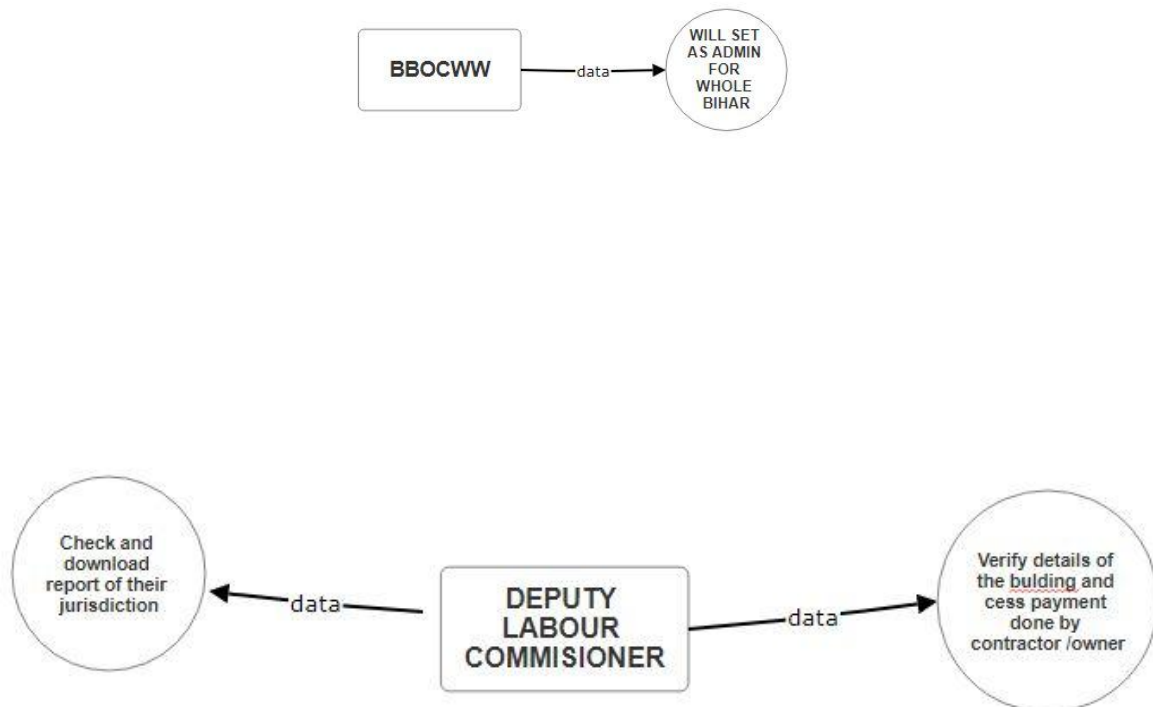
DFD 1 (Level 1.2)



DFD 1 (Level 1.3)



DFD 1 (Level 1.4)



Non-Functional Requirements

Performance Requirement

The web and mobile application will be used by every block of state of Bihar. The application may get slow down with the recommended hardware having concurrent users above 10,000 nos. The application may not work very old legacy phones and need time to time updating from play store. During third-party authentication failure, application may cause system integration and stability issue. On very low bandwidth internet service , the performance may degrade for the mobile application. Smart dashboard can be customized and based on query fetching response time the dashboard may have flickering issues.

Safety Requirement

There are safety measures to avoid any type of security hazards. The solution will have facilities for tamper proof data storage to avoid unauthorized data tampering. All hosting platforms must have the internal security majors at infrastructure level to avoid data crash , tampering or data corrupt issues.

There will be a password policy for creating and maintaining strong password. The solution will validate automatically passwords as per the password policies. The solution will enforce changing of the default password when the user first logs into the system. The system will be built on certain specified guidelines and under ISO 27001 security standards for application safety and security.

The database may get crashed at any certain time due to virus or operating system or hardware or network failure.

Therefore, it is required to take the database backup at regular interval of 1 hour.

The sensitive data will be saved in the encrypted format. Therefore, data will be in safe during transmission across the network. Also, database mirroring and clustering along with HA environment can be used as a safety requirement.

Security Requirement

Anyone developing a public Web site today really needs to have a strategy for protecting their site from digital terrorists. It is also very much important for BBCLWB project because it carry very much crucial and valuable data with this in mind, as a security-conscious organization. Release of quality with high availability & high-performance application are generally our main objective.

All the approval data transfer in the system will be transited in encrypted format and the application will have the web security feature that protect against URL Manipulation. It will ensure that the approved data is not changed / altered by anyone.

Authenticate user

User authentication will be in following mode.

- ✚ Desktop web Browsers

- ✚ Mobile web browsers

- ✚ Mobile App

Note: Lower version of web browsers will not be supported due to security concerns.

Role Based Access Control

- ✚ Role based access control will be created for different officials

- ✚ Assign only required modules to be available the officers

Authorized based on role member ship

- ✚ If LS/LEO/DLC logins in, they will get to see the users only based on their location or mapped to them.

Accessing Downstream Resource Manager with a Fixed Trusted Identity

Utilizing a fixed trusted identity for accessing downstream resource managers offers several advantages:

- ✚ **High Scalability:** Supports efficient connection pooling, enhancing system performance.

- ✚ **Minimal Administrative Effort:** Reduces the complexity of managing individual user access.
- ✚ **Simplified User Experience:** Users do not require direct access to downstream resources.

Data Validation

Ensuring **data access security** is crucial to prevent unauthorized access to the database. Every piece of data must undergo thorough validation before being submitted to the backend. The following measures will be implemented to maintain data integrity and security:

1. **Input Validation:** All data must be validated for length, format, and input constraints before processing.
2. **Regular Expressions:** Proper regular expressions should be used at the backend to enforce format validation.
3. **Query Security:** Dynamic query generation based on user input should be avoided to mitigate SQL injection risks. User-provided parameters must not be directly involved in query construction.
4. **Stored Procedures:** Wherever possible, stored procedures should be used instead of executing direct query batches to enhance security and maintain structured access control.
5. **Error Handling and Rollback:** In case of an error, proper rollback mechanisms must be implemented to prevent the database from entering an inconsistent state.
6. **Password Encryption:** The database should not store plaintext passwords. If passwords are required, they must be encrypted to ensure secure storage.

Other Security Feature

Data Encryption: Using Advanced encryption techniques for sensitive data like worker records, scheme details, and financial transactions.

Authentication: Implementing OAuth 2.0 or JWT for secure access control, including biometric authentication with Adhaar

Secure APIs: To Ensure all API integrations (e-Shram, Payment Gateways, SMS services) are secured with SSL and follow best practices for security.

Periodic Security Audits: Conducting Regular penetration testing and security audits will be performed to ensure the system remains resilient against cyber threats.

Except the above security details following check point will be there to ensure protect the web site from digital terrorist.

- ✚ Implement Strict Server side and Client-side Input Validation.
- ✚ Protect CSRF Attack
- ✚ Create Hidden field
- ✚ Generate random number using Randomize function
- ✚ Implement Strict Session Management.
- ✚ Implement solutions against brute force attack on login page.
- ✚ Strong password management
- ✚ Display custom error messages in case of an error.
- ✚ Protect Information leakage
- ✚ Protect OS command injections
- ✚ Exception Handling

Exception handling will be an important part of the security part for BBOCWB project. We will add a Global Application Error Handler in the application, and that will handle all the exceptions that are occurring in our application.

Create Log Files is the same class as we added at the starting for logging the errors. Apart from that we can write separate exception handling codes for each function, or wherever it is required. We can

add custom error code to configuration file to redirect the users to some error page rather than displaying the error.

Software Quality Attributes: Reliability & Reusability

Reliability

The application will incorporate robust web security features to ensure a reliable and uninterrupted user experience. It will be accessed securely via **HTTPS**, implementing the following security measures:

- ✚ **API-Driven Architecture:** All data transactions will be handled via APIs, ensuring complete encryption of traffic. This prevents unauthorized access and ensures data security.
- ✚ **Server Identity Verification:** The identity of the remote server will be authenticated using **digital certificates**, enhancing trust and preventing security threats.
- ✚ **Optional Client Authentication:** The system will support **client-side certificate authentication**, allowing secure logins using chip cards instead of (or alongside) traditional passwords.

These security measures will enhance system reliability by safeguarding user data and ensuring secure communication.

Reusability

The application will be developed using a modular approach, ensuring that components can be efficiently reused. Key aspects of reusability include:

- **Predefined Frameworks:** Certain components, such as the **admin console**, will leverage established frameworks for **rapid application development** while maintaining consistency.
- **UI/UX Alignment:** The new system's **user interface and experience (UI/UX)** will align with the existing portal layout, ensuring familiarity for users while integrating modern enhancements.

By reusing standardized components and frameworks, development efficiency will improve, reducing redundancy and ensuring a seamless user experience.

User & Developer Requirements

Portal and Mobile Application Functional Considerations

The **web admin panel** for the portal will be accessible exclusively through the web application, as administrative functions require a **laptop or PC** for efficient management. Given the complexity of administration tasks, a mobile interface will not be available for web admins.

The **chat box feature** will provide **limited support**, specifically assisting labourers with information regarding welfare schemes and general inquiries. It will be an **FAQ-based system**, offering pre-defined responses rather than real-time conversational support. This ensures quick access to relevant information without requiring extensive backend processing.

The **website will incorporate a moderate Content Management System (CMS)**, allowing the web administrator to update and manage dynamic content for public viewing. This will enable real-time modifications to scheme details, announcements, and other essential information.

For the **mobile application**, users must update the app manually upon each new release to ensure they have the latest features, security patches, and performance improvements. The **portal and mobile application will be device-independent**, ensuring seamless functionality across different screen sizes and operating systems. The system is designed with a **responsive user interface**, ensuring compatibility with desktops, tablets, and mobile devices without compromising user experience.

User authentication will be secured through **OTP and Aadhaar-based verification**. However, in cases where external authentication servers (such as Aadhaar or OTP gateways) experience downtime or technical issues, users may face temporary access restrictions.

The portal will support **bilingual translation**, but this feature will be managed manually via the web admin panel. Content translation will not be automated but rather dependent on the **admin's input values** in the backend system.

To optimize performance and storage efficiency, document uploads will be subject to **file size limitations**. Users will be required to **compress large documents** before uploading them to the system to ensure smooth data handling and prevent excessive server load.

Glossary , Assumptions and Dependencies

SL#	Term	Definition
1	BBCLWB /BOCW/BBOWB	Bihar Building and Other Construction s Welfare Board
2	MIS	Management information System
3	RFP	Request for proposal
4	MOM	Minutes of Meeting
5	SA	System Administrator
6	DLC	Deputy Worker Commissioner
7	LEO	Labour Enforcement Officer
8	UPS	Undistributed Power Supply
9	OS	Operating System
10	TCP	Transmission Control Protocol
11	SMS	Short Message Service
12	URL	Universal Resource Locator
13	CSRF	Cross-Site Request Forgery
14	SQL	Structured Query Language

Appendix A: Project Plan

Milestones	Tasks	Deliverables	Week 1	Week 3	Week 6	Week 9	Week 12	Week 15	Week 18	Week 21	Week 24	O&M 5 Years	
Phase 1: Development and Deployment													
Milestone 1	SI Resource Mobilization	Finalized Project Plan Finalized Requirement Report	W1	W1									
Milestone 2	Requirement Gathering & SRS	Approved System Requirement Specification (SRS)		W3	W3								
Milestone 3	Approval of SRS				W4								
Milestone 4	Design and Development and Release of Beta Version	Successful UAT Report			W6	W9	W9						
Milestone 5	UAT and Bug Fixing						W12	W12					
Milestone 6	Security Audit	Sign off Certificate						W15	W15				
Milestone 7	Application Deployment on SDC & Go-Live								W18	W18	W20		
Milestone 8	Completion of Training	Training completion certificate from the client Project documents mentioned under subsection 7.2.09									W21	W24	W24
Phase 2: Operations and Maintenance													
Milestone 9	Maintenance Support and Change Management	SLA adherence report on quarterly basis											O&M
Milestone 10	Helpdesk Support												O&M

Appendix B: Man-Hours Plan

Manpower	Week 1		Week 3		Week 6		Week 9		Week 12		Week 15		Week 18		Week 21		Week 24		O&M 5 Years
Project Manager	W1	W1	W3	W3	W6	W6	W9	W9	W12	W12	W15	W15	W18	W18	W21	W21	W24	W24	O&M
Business Analyst	W1	W1	W3	W3	W4														
Database Administrator					W6	W6	W9	W9	W12	W12	W15	W15	W18	W18	W20				
Development Lead					W6	W6	W9	W9	W12	W12	W15	W15	W18	W18	W20				
Developer					W6	W6	W9	W9	W12	W12	W15	W15	W18	W18	W20				
Testing Lead									W12	W12	W15	W15	W18	W18	W20				
Tester									W12	W12	W15	W15	W18	W18	W20				
Network Expert																W21	W24	W24	O&M

Appendix C: API Integration requirement

- **Adhaar Integration:** For biometric authentication.
- **UPI Payment Integration & NACH Integration:** For payments.
- **E-Shram Integration:** For exchanging worker data with the central government.
- **SMS Integration:** For OTP authentication
- **SSL:** Secured socket layer integration for security